

Universiteit Leiden
BSc Data Science and AI

Modeling Humor in Human-Centric AI: A Holistic Approach to Humor Detection and Interpretation

Xocas Álvarez Jorge

[S3443752]

Supervisors:
Dr. Max van Duijn
Lennard Froma

BACHELOR THESIS

Leiden Institute of Advanced Computer Science (LIACS)
www.liacs.leidenuniv.nl

1/06/2025

Abstract

The recent evolution of Large Language Models (LLMs) has opened new avenues for exploring their capacity to emulate human-like and human-centric behaviors. This thesis investigates how several leading LLMs perform in understanding and interpreting humor. This study aims to investigate in a holistic manner, both qualitatively and quantitatively, how effectively LLMs can detect, analyze, and interpret text-based humor. The ColBERT dataset, created by Cornell researchers, is employed alongside a self-collected corpus of humorous text derived from Reddit, spread across several categories and subcategories of humor. Both datasets exhibit a 50/50 split of humorous and nonhumorous content. First, the various LLMs are asked to identify humor, along with a confidence rating. Secondly, the various LLMs were asked to interpret humor, assigning scores across several categories, such as, but not limited to, appropriateness, offensiveness, and funniness. Apart from this, they were asked to explain jokes selected through random distribution from the identification task, such as false positives, false negatives, or mismatches between humor predictions and their personal opinion.

The findings indicate overall that LLMs are largely proficient in detecting linguistic and contextual humor, such as puns and wordplay, best explained by Incongruent Humor Theory. On the other hand, all the LLMs struggled with niche, regional, and emotionally driven humor (e.g., NSFW or social dynamics), particularly those requiring ToM or cultural context, as explained by Superiority and Relief theories. On an individual level, Grok outperformed all LLMs, achieving the highest accuracy, with Llama trailing close behind. ChatGPT had a middling accuracy, and Deepseek performed the worst. The study also demonstrated the comedic preferences of the LLMs. Mismatches between classification and 'perceived' funniness often stem from low-quality or unoriginal jokes, with offensiveness raising the quality threshold for humor perception. Each LLM further had its own criteria for perceived funniness and humor detection. Upon inspection of the various variable rankings, Deepseek and ChatGPT emphasized offensiveness and appropriateness, while Grok and Llama focused on funniness and confidence.

Contents

1 Introduction **1**

2 Background and Related Work **1**
2.1 What is a Large Language Model . . . 2
2.2 Humor Theory . . . 2
2.2.1 Incongruity Theory . . . 2
2.2.2 Superiority Theory . . . 3
2.2.3 Relief Theory . . . 3
2.2.4 Benign Violation Theory . . . 3
2.3 Implications for LLM research . . . 4
2.4 Previous Computational Approaches to Humor . . . 4
2.4.1 Humor Classification . . . 4
2.4.2 Humor Detection . . . 5
2.5 Further Motivation/knowledge gap . . . 5

3 Research Question **6**
3.1 Hypothesis . . . 6

4 Methodology **7**
4.1 Datasets . . . 7
4.1.1 ColBERT Database . . . 7
4.1.2 Custom Reddit Database . . . 7
4.2 Selection and Motivation of Large Language Models . . . 8
4.2.1 Chosen Large Language Models . . . 8
4.3 Tier 1: Humor Detection . . . 9
4.4 Tier 2: Humor Explanation . . . 10
4.4.1 Quantitative Analysis . . . 10
4.4.2 Qualitative Analysis . . . 11
4.5 Methodology for Analysis . . . 11
4.5.1 Quantitative Analysis . . . 12
4.5.2 Qualitative Analysis . . . 12

5 Results **12**
5.1 Reddit Data Set Results . . . 12
5.1.1 Tier 1 Results . . . 13
5.1.2 Tier 2 Results . . . 14
5.2 ColBERT Data set Results . . . 16
5.2.1 Tier 1 . . . 16
5.2.2 Tier 2 . . . 16

6 Discussion **17**
6.1 Discussion of Inferential Statistics and Preliminary Results . . . 17
6.1.1 Reddit Dataset . . . 17
6.1.2 ColBERT Dataset . . . 19

6.2 Humor Mismatch . . . 20
6.2.1 Quantitative Analysis . . . 20
6.2.2 Qualitative Analysis . . . 24
6.3 Variable Correlations . . . 25
6.3.1 Classification Predictors. . . . 25
6.3.2 High and Low Funniness Predictors . . . 27
6.3.3 Overall Correlations . . . 29
6.4 Analysis by Category . . . 31
6.5 Limitations, Weaknesses and Areas of further Improvement . . . 32
6.5.1 Methodological Limitations . . . 32
6.5.2 Data Set Weakness . . . 33
6.5.3 LLM related issues. . . . 33

7 Areas of Further Research **34**

8 Conclusion **35**

9 Appendix **37**

10 Results **37**
10.0.1 Qualitative mismatch samples . . . 40

References **43**

1 Introduction

Humor is a complex cognitive process, requiring an understanding of sarcasm, irony, timing, cultural references, and implicit social cues [3]. Given the complexity and variety of humor, AI systems and humans alike can struggle in identifying humor, especially more niche, recent, or online humorous content, such as dark humor or memes. Humor is not a universal standard; it varies across cultures, standards, and different contexts, thus presenting an intriguing challenge for an LLM to tackle.

Humor is a cornerstone of human communication, facilitating social bonding, entertainment, or transmitting messages, beliefs, or motifs [20]. In human-centered AI, the ability of an LLM to engage with humor effectively is a measure of its progress and alignment with human-style communication. Although recent advances have demonstrated the remarkable proficiency of LLM in natural language tasks [12], their performance in humor-related tasks remains understudied. This is especially true given a near-complete lack of studies approaching humor and LLMs from a holistic perspective.

Furthermore, while humor detection requires the integration of linguistic, cultural, and social cues, it often requires a form of theory of mind (ToM) to infer contextual intent [6, 3, 18]. Although LLMs appear to perform well on ToM tasks, some evidence suggests that their success may stem from pattern recognition in training data, raising questions about genuine understanding, similar to Searle's Chinese Room argument [30]. By examining how LLMs interpret and explain various styles of humor, one can likely gain insight not only into how they perform with humor but also the reasons why while also approaching a ToM task from a novel approach.

Addressing such a gap is crucial to developing human-centered AI systems that are effective in various social contexts and align with and understand human communicative intent. This is especially emphasized by the increasing integration of AI systems and LLMs into human-centric and adjacent tasks. Addressing such a gap probably has important practical applications, be it virtual assistants, content moderation, or improving human-AI interaction across several fields. In these tasks, understanding and addressing humor, whether it is funny or not, will be significant.

This bachelor's thesis, conducted at the Leiden Institute of Advanced Computer Science (LIACS) under the supervision of Dr. Max van Duijn and Lennard Froma, investigates the ability of various LLMs to detect, interpret, and explain contemporary humor. The motivation behind this study is two-fold. The first being to address the current limited research gap on humor, and secondly to gain a holistic understanding of their approach to humor, and by extension, gain further understanding into the alignment of LLMs with the ability to reason and work with human humor and communication. By evaluating the performance of LLMs in diverse humor types, this study seeks to delineate their strengths, identify limitations, and propose directions for future research in human-centered AI.

2 Background and Related Work

LLMs have been demonstrated to be effective in text generation and sentiment classification [12, 1], however, their ability to do so with certain types of humor, and especially more contemporary or niche humor, remains understudied. Even in current studies addressing LLMs and humor, such an

approach is not holistic, but rather focuses only on detection, classification, or another individual piece rather than a comprehensive approach across detection, classification, and explanation. By first examining current literature, this study will gain the necessary information to formulate a robust study and explicitly define and identify the current knowledge gap regarding LLMs and humor.

2.1 What is a Large Language Model

A Large Language Model or LLM is a type of AI model that is designed to process and generate human-like text. It is trained on huge datasets of text from the internet, books, or other sources to perform tasks such as answering questions, classifying text, or generating appropriate content by predicting and producing sequences of words based on patterns learned during training. While LLMs excel in an array of natural language processing tasks, they may struggle with more complex tasks that require true contextual understanding rather than relying on statistical patterns [9]. In later sections, selected LLMs will be introduced.

2.2 Humor Theory

When interpreting the performance of LLMS in the outlined humor tasks, a theoretical framework which we can use to asses and interpret results is important. This section brings in concepts from humor which are relevant to the various categories of humor this study asses. With this, we have a lens through which we can focus and interpret results. For this study, we can define humor to be something which may be comical or funny for at least a certain subset of a general audience or public.

2.2.1 Incongruity Theory

Incongruity theory stipulates that humor arises from the perception of unexpected or contradictory elements within a text, video, or whichever vehicle of delivery for humor [4]. The theory underscores a mechanism nu which a surprise created by mismatching realities and expectations leads to laughter. In the context of this study, incongruity theory aligns with several subreddits. Reddit is a social media platform organized into communities called subreddits, each focused on a specific topic or theme. Users can post and comment within these subreddits, making it a rich source of user-generated content and internet culture. Take r/puns for instance, where humor stems from unexpected word play, essentially exploiting multiple meanings of a word when another is expected. Other subcategories are situational absurdity, where incongruent situations and events may create humor. Other categories may rely on this mechanism to different extents, such as but not limited to Emoji Absurdism (Visual Humor) and Anachronism (Temporal Humor), which also relies on incongruity through exaggerated or time-displaced elements [16].

For LLMs, incongruity theory informs us of what may be required to interpret humor. This study has two levels of analysis. These will be further developed in later sections. For example, in level 1, an LLM must identify incongruous elements (for example, detecting the dual meaning of a pun) to classify a text as humorous or not. In Tier 2, this may be understanding the unexpected twist of a joke to assign a ranking of funniness or clarity. However, detection of incongruity may be

obfuscated by subtle nuances, requiring cultural context, such as in Regional/Subcultural humor, where localized knowledge is needed [39].

2.2.2 Superiority Theory

Superiority Theory posits that humor can emerge from a sense of superiority, take for instance, mockery or schadenfreude [23]. One may find this humorous as it elevates the own self-esteem. While not exhaustive of all humor, it provides an interesting framework for the analysis across several categories of humor. This can be directly tied to several categories, subcategories and subreddits such as but not limited to like Authority Mockery (Power Reversals), where humor targets authority figures (e.g., `r/okboomer` posts mocking older generations), and Schadenfreude subcategories such as Misfortune and Cringe Comedy, where humor arises from others' failures or awkwardness (e.g., `r/leopardsatemyface`, `r/sadcringe`).

For LLMs, this would require them to understand social hierarchies, which are very much culturally dependent. In Tier 1, LLMs must identify jokes as humorous rather than simple, cruel insults, but a joke can fall in both categories. In Tier 2, this framework can be used for quantifying categories such as offensiveness or appropriateness. This may further require some ToM in inferring power dynamics and potential harm [7]. Here, LLMs may struggle with the ethical nuances [16] and likely will vary significantly based on content moderation policies and training data.

2.2.3 Relief Theory

Relief Theory states that humor serves as a release of psychological tension or suppressed emotions [23]. This can often be in response to taboo or stressful topics and situations [23]. From this view, humor can be an outlet for addressing societal constraints. Relevant categories, subcategories, and subreddits could be as follows, but not limited to: Gallows Humor and Terminal Illness (Morbid Humor) or Taboo Topics such as Religious and Political Incorrectness, where humor violates social norms (e.g., `r/exchristianmemes`, `r/offensivememes`).

Within the context of this study and theory, LLMs cannot have psychological tension or suppressed emotion [29]. However, the various humor theories can serve as a theoretical framework to asses how LLMs perform or can also serve as a ToM test[7]. Training data may not be enough to ensure good performance on humor identification, where Relief theory may play a large role. In Tiers 1 and 2, the tension release will be critical in identifying jokes and ranking their appropriateness and offensiveness. A further cultural component will further test LLMs. They may face challenges in balancing cultural sensitivity with humorous intent in classifying humor, such as in subcategories like political incorrectness. Here, the relief effect can be polarizing [16].

2.2.4 Benign Violation Theory

Benign Violation Theory or BVT offers a framework where humor arises from violating a social, moral, or physical norm while being perceived as benign or safe [21]. BVT differs from relief in that relief acts to dissuade tension or anxiety, while BVT does not necessarily require such tension or anxiety in the first place. BVT stipulates three conditions for humor:

- A norm is violated (e.g., a taboo topic is broached)
- The violation is non-threatening or harmless
- Both the violation and benignity are recognized concurrently

This theory is helpful for humor that may push political, social, emotional, etc barriers. Several LLMs may struggle with this, such as DeepSeek, especially with political humor that may find itself at odds with its moderation policy and cultural context. While trained on western data, the blockage of western social media in China without a VPN may impact its abilities. Furthermore, political correctness standards in China do limit the content Deepseek will interact with, especially those of a political nature and critical of the Chinese Communist Party or China overall. [38]. Hence, this provides a lens which to interpret LLM performance across various data sets.

BVT can be directly applied to several of the Reddit datasets, such as NSFW humor, taboo topics (religious and political incorrectness), or r/offensivememes. Morbid humor or the mockery of Authority may also be potentially analyzed with a BVT framework as it violates social hierarchies in a humorous, non-threatening way [21]. In Tier 2, BVT could be used to explain choices in ranking Offensiveness and appropriateness or interpret explanations given for why a certain text may be funny or not.

2.3 Implications for LLM research

The theories outlined above give us a framework to evaluate LLM performance in humor tasks, outlining potential difficulties and challenges, and steps in humor detection. This can allow us to understand not only how LLMs perform the way they do, but more importantly, attempt to provide an explanation why and interpret results, both qualitative and quantitative. Tier 1 and especially Tier 2 will test such theories in seeing how LLMs classify and interpret humor.

2.4 Previous Computational Approaches to Humor

This literature review examines recent studies from within the previous decade relevant to this study. This will be broken down into two levels of studies. The first level will be a review of studies focusing on humor detection, while the second Tier focuses on humor analysis (explaining why text may or may not be funny).

2.4.1 Humor Classification

Humor classification, often framed as a sentiment analysis task, involves labeling text as humorous or non-humorous [26]. A meta-analysis by Radev and Mihalcea [26] notes that many studies rely on pre-labeled datasets, such as memes or social media posts, limiting LLMs' ability to autonomously detect humor. For instance, Jentzsch and Kersting [19] found that ChatGPT achieves 78% accuracy in classifying simple jokes but struggles with irony or non-English humor. Similarly, Vaz Valois, Romanowski, and Tso [35] evaluated ChatGPT, Claude, and DeepSeek on stand-up comedy transcripts, reporting a 51% accuracy compared to human annotators' 41%, highlighting

challenges with contextual humor like dark humor or satire. These findings suggest that LLMs' performance often stems from training data patterns rather than a specialized humor detection capability [30]. Research on newer LLMs, such as Grok or DeepSeek, and niche humor types (e.g., internet memes) remains limited, creating a knowledge gap. This gap is critical, as complex humor types elicit unique challenges for LLMs, necessitating further investigation into their classification abilities across diverse genres.

2.4.2 Humor Detection

Humor analysis, which involves interpreting and explaining why text is humorous or not, extends beyond classification and remains underexplored for large language models (LLMs) [20]. Unlike detection, analysis requires understanding cultural, social, and contextual cues, often necessitating a ToM to infer intent [6, 18]. Jentzsch and Kersting [19] found that ChatGPT struggles to explain invalid jokes, reflecting limitations in contextual reasoning.Jentzsch and Kersting [19] found that ChatGPT struggles to explain invalid jokes, reflecting limitations in contextual reasoning. That is jokes that lack a recognizable setup, do not meet humor expectations (e.g., they're incoherent, poorly worded, or just not funny) amount other reasons. Similarly, Vaz Valois, Romanowski, and Tso [35] noted that ChatGPT and DeepSeek falter in interpreting stand-up comedy punchlines due to cultural nuances. Popova and Dadic [25] reported that GPT-based models poorly explain puns in multilingual contexts, while Chaudhary, Goel, and Mamidi [14] highlighted LLMs' challenges with conversational humor's dynamic cues. Subjective humor perception, particularly for niche forms like dark humor or memes, remains a bottleneck, as LLMs struggle to align with human judgments of funniness or content moderation policies [37]. The latter will likely require testing to subvert and extract a response.

2.5 Further Motivation/knowledge gap

LLM Existing studies often focus on a single task, such as classification [10], evaluation [17], or generation [19]. Current literature does not approach such an issue in a holistic manner, leaving out much information when wishing to study LLMs from a Human AI-centric lens. Furthermore, such studies mostly do not compare LLMs against each other, often focusing on custom-built LLMs for said task or a singular task rather than a comprehensive approach. Meta-analyses underscore this fragmentation, noting that computational humor research prioritizes isolated tasks (e.g., detection, generation) over holistic approaches [26, 37].

Given the recent emergence and significant impact of newer large language models (LLMs) like Grok and DeepSeek, addressing the limited research on their humor-processing capabilities is critical for advancing human-centered AI [12]. This knowledge gap, particularly in understanding how these models detect, interpret, and explain contemporary humor such as dark humor and memes, underscores the need for a holistic approach [26]. The following section formulates a research question to guide this investigation and bridge the identified gap in the scientific literature.

The current fragmentation in academic literature limits our ability to assess the broader communicative competence of LLMs, especially in human-centric applications where understanding nuance, emotion, and context is vital. More holistic approaches can better illuminate how LLMs

perform outside isolated tasks and can be asses in how they mirror or diverge from human cognition. This opens the door for discoveries of how LLMs align with Theory of Mind reasoning, improving cross-cultural robustness, and informing safer, context-aware content moderation strategies.

3 Research Question

The research question of the study can be defined as follows:

How can Large Language Models Detect, Interpret, and Explain Humorous Content?

With the following guiding questions to further refine and focus the scope of the study:

- What are the differences between Grok, DeepSeek, and ChatGPT in their ability to perform this task?
- How do these large language models perform across various types of humor?

3.1 Hypothesis

Given this research question, a hypothesis can be formulated. Based on the various LLMs and Comedic categories, it is likely that performance will vary specifically with the humor category and the specific LLM. A further breakdown is as follows:

- **Grok**; This LLM will likely perform best, given its training background. It likely will best detect and interpret, especially NSFW humor, given its more relaxed content moderation policy, and will likely find such jokes more humorous.
- **DeepSeek**: DeepSeek is likely to be the worst performing in this task, given the English and likely Western-centric datasets employed, contrasting with DeepSeek's unique cultural background. While trained ostensibly on Western data, much of Western social media is banned within mainland China, likely impacting its training. However, the extent of this is unknown. It will likely struggle with inappropriate forms of humor, especially if it falls outside acceptable norms for its cultural context.
- **ChatGPT**: ChatGPT will likely perform well with the exception of non-appropriate forms of humor. This can be attributed to its content moderation policy. Given that ChatGPT is the most established LLM and market leader, it will likely perform well [2].
- **Llama**: Llama will likely perform well in Linguistic, Technical, and Structural humor, given its academic background and strength in pattern recognition. However, it is unknown how Llama will perform, as while academically based, its development by social media giant Meta may allow it to perform well on non-traditional humor.

4 Methodology

This study employs a two-Tiered framework to evaluate the humor proficiency of LLMs, including Deepseek, Grok, ChatGPT, and Llama, across detection and analysis tasks. This study will employ two datasets: a custom-made Reddit dataset and the ColBERT Humor Detection corpus. These data sets consist of humorous context mixed with non-humorous content. These data sets are fed to each LLM in two Python scripts: one for Tier 1 (`process_humor_datasets.py`) and another for Tier 2(`Tier_2_analysis.py`). Below is a detailed explanation of how the datasets are created, how data processing works, the LLM configurations, sampling strategy, analysis, and other relevant setup details.

4.1 Datasets

4.1.1 ColBERT Database

The first dataset employed is the ColBERT dataset, developed by researchers at Cornell University. This dataset consists of 200,000 entries, of which half are humorous and half are not. The data set was custom-built for LLM humor detection, addressing some lags in prior data sets [**annamoradnejad2020ColBERT**]. The humorous content includes one-liners and puns that come from Reddit and various web pages. The dataset was enforced with high cleaning standards, achieved by eliminating duplicates, an advanced parsing system, and by rejecting invalid inputs. Invalid inputs represented less than one percent of the dataset. These texts are purposely designed to test LLMs in detecting linguistic incongruities and script oppositions, aligning with the established humor frameworks [4, 27]. Non-humorous content originates from factual texts, news headlines, Wikipedia excerpts, and general knowledge snippets.

The ColBERT data set is useful given its custom purpose-built corpus, along with its short self contained texts. Another strength is its size, which can allow for robust analysis. It is limited, however, in that it does not contain any other labels aside from true/false for humor, meaning analysis would be limited to general insights. Regardless, this dataset proves to be effective for a general humor category analysis.

4.1.2 Custom Reddit Database

The second dataset used is a custom-made Reddit Humor Dataset, consisting of humorous texts scraped from Reddit and several non-humorous texts from Wikipedia. These are standardized to be of similar lengths. The dataset is organized into seven primary humor categories, each with specific subcategories, as defined in the `SUBREDDIT_MAPPING` structure of the dataset creation script. The categories and subcategories can be viewed in Figure 1. Definitions for categories, subcategories, and subreddits can be viewed in the appendix. The composition of the categories and subcategories was done by browsing through and determining subreddits that fit most effectively into said category 10.

The dataset was constructed using the Python Reddit API Wrapper (RAW), which combined the header and body of only text-based posts. Also scraped were upvotes, downvotes, and the score of the post (being upvotes minus downvotes. Posts were filtered to only include those with a minimum

score of 3, lengths between 20 and 450 characters, and no embedded media (e.g., images, videos) to ensure compatibility with LLM text processing. Then, each subreddit is supplemented with an equal number of non-humorous entries from Wikipedia with length constraints (20–450 characters). These can generally be assumed to be humor-free. With approximately 20,000 entries, this dataset allows for a deeper analysis and breakdown of LLMs and humor. However, it may face challenges in the form of quality and cleaning when compared to ColBERT, such as but not limited to: informal language, user-generated inconsistencies, or residual noise.

Figure 1: Humor Categories and Reddit Communities for the Reddit Humor Dataset

Category	Subcategory	Reddit Communities
Linguistic Humor	Wordplay (Puns, Homophonic, Paraprosdokians)	r/puns, r/dadjokes, r/wordplay, r/boneappletea
	Semantic Shifts (Double Entendres, Malapropisms)	r/dirtyjokes, r/innuendos, r/lostintranslation
Contextual Humor	Situational Absurdity (Surreal, Anti-jokes)	r/surrealhumor, r/antijokes, r/dryhumor
	Cultural References (Historical Satire, Pop Culture Parody)	r/historymemes, r/popcultureparody, r/memes
NSFW Humor	Morbid Humor (Gallows, Terminal Illness)	r/darkjokes, r/morbidhumor, r/toosoon
	Taboo Topics (Religious, Political Incorrectness)	r/exchristianmemes, r/offensivememes, r/imgoingtohellforthis
	Existential Nihilism (Absurdist Dread, Cosmic Horror)	r/absurdism, r/nihilistmemes, r/cthulhumemes
Social Dynamics	Power Reversals (Underdog, Authority Mockery)	r/wholesomememes, r/antiworkmemes, r/genzhumor
	Schadenfreude (Misfortune, Cringe Comedy)	r/instantkarma, r/cringe, r/sadcringe
Technical/Structural	Pattern Interrupts (Rule-of-three, Misdirection)	r/unexpected, r/plotwistmemes, r/nonononoyes
	Visual Humor (Emoji Absurdism, Meta-Humor)	r/emojipasta, r/deepfriedmemes, r/metamemes
Regional/Subcultural	Dialect Humor (British, Southern US)	r/britishproblems, r/holdmybeer
	Nerd Culture (Programming, Science Puns)	r/programmerhumor, r/chemistrymemes, r/techhumor
Experimental	AI-Specific Challenges (Sarcasm, Ethical Edge Cases)	r/sarcasm, r/unethicallifeprotips, r/ironicmemes
	Temporal Humor (Anachronism, Future Shock)	r/timetravelercaught, r/cyberpunkmemes, r/futuristmemes

Note: Representative subcategories and communities listed; full mappings in appendix.

4.2 Selection and Motivation of Large Language Models

To comprehensively evaluate the humor proficiency of Large Language Models (LLMs) in detection, interpretation, and explanation tasks, this study selected four LLMs: Grok, DeepSeek, ChatGPT, and Llama. These models were chosen for their diversity in design, training data, cultural influences, and organizational backgrounds, which allows this study to have a robust framework when analyzing performance across various humor types and contexts. This diversity is critical to understand different training philosophies and data, and how this can impact how LLMs' engage with the complex, human-centric task of humor processing.

4.2.1 Chosen Large Language Models

- **Grok 3**: Grok is developed by XAI, being a transformer-based model, optimizing for what XAI claims to be *"truth-seeking and conversational reasoning"* [33]. The model is proprietary, meaning one cannot view the weights and architecture of the model. It is, however, known that Grok is trained on a diverse web corpus, including X (formerly twitter), where a focus is contextual understanding. This ideally should make it suited for understanding (more niche

areas of) humor. Grok has also proven more flexible in accepting and interacting with content that may not reach safe for work standards [31]. Given its design, it seems poised to excel in processing nuanced or contextual texts, such as puns or cultural references. This means it may likely be able to perform well in Incongruity and Script Opposition theories by detecting linguistic surprises and semantic shifts[4, 27]. This LLM was chosen for its unique training and flexibility, likely to yield interesting results.

- **DeepSeek**: DeepSeek is a relatively newer model, realized in the first quarter of 2025, and well-performing across a variety of metrics [32]. It is trained on a corpus of web texts and code, while containing 671 billion parameters, compared to Grok and ChatGPT, which contain 314 and 175 billion respectively. Deepseek also originates from China, and is subject to cultural limits, such as a refusal to discuss political issues that may be prohibited in China [**source**]. This means it may also struggle with more cultural or niche jokes. This LLM was chosen to provide an open source and distinct cultural perspective of an LLM to judge and compare to others.

- **ChatGPT**: OpenAI's GPT-3.5 was chosen as a neutral standard. With the first iteration of Chat-GPT being released in November of 2022, it is the oldest and most established LLM in the market, excelling in general-purpose language tasks specifically [13]. Its training has consisted of not only web texts but also books and user interactions. However, it remains the least flexible given its more stringent moderation policy, which may impact its ability to interpret humor such as dark humor or political incorrectness [16]. It was chosen because of its widespread usage as the most popular LLM at the time. [1]

- **Llama**: Llama-3.3 is developed by Meta AI and accessed via the Together API tool. Llama is the model with the smallest number of parameters, with only 70 billion parameters. It is optimized primarily for research tasks. [34]. Llama is primarily trained on web texts and academic corpora, which may hinder performance when analyzing humor, especially niche or contextual parts. Llama's research focus may limit ToM capabilities for intent-driven humor compared to conversational models [39]. However, given its instruction tuning, it will likely perform somewhat well in structured tasks like scoring and explanation. This model was chosen given its open-model nature, where its architecture is known, and for its distinct training background.

4.3 Tier 1: Humor Detection

In Tier 1, large language models (LLMs) perform a binary classification task on texts from the Reddit and ColBERT datasets. For each text, LLMs provide three outputs:

1. **Classification**: A binary judgment (YES for humorous, NO for non-humorous).

2. **Confidence Score**: A numerical score (1–10, where 1 is least confident and 10 is most confident).

[1] By early 2025, nearly 86 percent of all tokens processed by LLMs were via ChatGPT, though newer models like GPT-4o have become the default in many applications [**HelpNet2025**].

3. **Personal Funniness**: A subjective opinion (YES/NO) indicating whether the model finds the text personally funny. Of course, an LLM cannot possess selfhood or emotions and hence a personal subjective opinion. However, this question serves to probe model behavior: it reveals how LLMs distinguish between recognizing a joke structurally versus finding it affectively engaging, offering insight into potential humor preferences or moderation constraints.

This is done with the `LLMAPI integration.py` script. For each LLM, each subreddit is processed one by one, in addition to the entirety of the ColBERT dataset, and stored as CSVs for each subreddit and ColBERT. These contain the previously listed categories as well as downvotes, upvotes, and score. The process is executed in batches to conform to rate limits and data quality. Invalid responses are rejected, and an advanced parsing system is used to extract values from invalid responses provided the values asked for are given.

This gives us the benchmark for humor identification, where the humor framework can identify a 'humor pathway' in which an LLM can detect a joke [4, 27]. Take Puns, for instance, in which unexpected wordplay has to be interpreted in a humorous way, or how often semantic shifts in dark humor can test script opposition. By collecting metrics, this study can quantify how LLMs perform with humor and, in turn, explain results through a humor theory framework. This can be used for asses the reasons for performance and areas of improvement.

4.4 Tier 2: Humor Explanation

The second Tier of evaluation investigates the capabilities of LLMs in interpreting humor. This is done in both a quantitative and qualitative manner.

4.4.1 Quantitative Analysis

Here, LLMs assign scores (1–10, where 1 is lowest and 10 is highest) to the following attributes, designed to capture different areas of humor perception:

- **Funniness**: Subjective appeal of the humor, reflecting alignment with human comedic preferences.
- **Offensiveness**: Potential to cause offense, critical for evaluating sensitivity in NSFW humor categories such as Political Incorrectness.
- **Originality**: Novelty and creativity of the humor, relevant for meme-based or experimental humor.
- **Appropriateness**: Suitability for general audiences, testing cultural and contextual awareness.
  Clarity: Linguistic comprehensibility, essential for detecting script oppositions in puns and one-liners.

Outputted is a CSV file containing all the previous parameters of Tier 1, while adding the previously mentioned 5 parameters to the new CSV.
From a technical perspective, the scoring process is done in batches to comply with rate limits and ensure quality responses. This allows us to quantify humor attributes, aligning with the humor

theory framework (Section **??**).These variables can be used to draw inferences about categories or potential humor pathways used for identification. Originality can signal more unexpected shifts (leading to greater humor) or ToM to infer cultural nuances and comedic intent, particularly in dark humor subcategories like Gallows Humor [4, 27, 7, 39]. By aligning metrics with humor theory, we have a robust framework for interpretation.

4.4.2 Qualitative Analysis

In the qualitative phase, we select samples to then feed to LLMs to explain why or why not they are humorous. This probes how the various LLMs interact with humor and allows for a more open interpretation. 40 samples are collected per LLM across eight categories to ensure a balanced selection for investigation. This is as follows:

- **True Positive (TP)**: Correctly classified as humorous (`true_label=True`, `LLM_classification=YES`), e.g., accurately detected puns or one-liners.
- **True Negative (TN)**: Correctly classified as non-humorous (`true_label=False`, `LLM_classification=N`) e.g., factual Wikipedia excerpts.
- **False Positive (FP)**: Incorrectly classified as humorous (`true_label=False`, `LLM_classification=YES`) indicating over-detection errors in non-humorous texts.
- **False Negative (FN)**: Incorrectly classified as non-humorous (`true_label=True`, `LLM_classification=`) indicating missed humor in comedic texts.
- **High Confidence**: Confidence score 7, reflecting strong model certainty in classification, useful for analyzing robust humor detection.
- **Low Confidence**: Confidence score 3, indicating uncertainty or ambiguity, highlighting challenging cases.
- **Funny/Not Personally Funny**: Classified as humorous but not personally funny (`LLM_classification=` `LLM_personal_funny=NO`), e.g., recognizing technical humor without subjective appreciation.
- **Not Funny/Personally Funny**: Classified as non-humorous but personally funny (`LLM_classification` `LLM_personal_funny=YES`), e.g., finding subjective humor in a non-comedic text.

From a technical perspective, the function aims to collect five samples per category, prioritizing diversity across humorous categories and subcategories as defined by the `SUBREDDIT_MAPPING`. To ensure broad representation, samples are grouped by category and subcategory and capped at 40 per LLM. Here, while not structured to any specific humor theories, the open-ended nature of this sub-phase allows the study to potentially use the entirety of the framework for analysis.

4.5 Methodology for Analysis

The methodology for the Analysis is as follows. Initial analysis will mainly be inferential statistics of both datasets, and will involve describing any patterns within them. After this, there will be a quantitative and qualitative analysis, which is as follows:

4.5.1 Quantitative Analysis

- **Accuracy Metrics**: For Tier 1, binary classification accuracy (YES/NO) was computed for each LLM on Reddit and ColBERT datasets, comparing `LLM_classification` to `true_label` [**p14**]. Overall accuracy and category-specific accuracy (e.g., Linguistic, NSFW) were aggregated across subcategories defined in `SUBREDDIT_MAPPING` [**p12**], using `quanti_analysis.py`. Mismatch rates between `LLM_classification` and `LLM_personal_funny` were calculated to assess subjective humor perception [**p17**].

- **Point-Biserial Correlations**: To identify predictors of humor classification and funniness, point-biserial correlations were computed using `scipy.stats.pointbiserialr` [**p21**]. For Yes/No classification, correlations assessed relationships between `Yes_No` (1=YES, 0=NO) and Tier 2 attributes (funniness, offensiveness, originality, appropriateness, clarity, confidence) [**p21**]. For High/Low Funniness, correlations used binary variables (`funniness` ¿ 7, ¡ 3) against other attributes, excluding funniness for High/Low analyses [**p22**]. Results were visualized as heatmaps (Figures **??**–**??**, [**p21**–**p23**]).

- **Random Forest Feature Importance**: To determine predictors of humor mismatches (`LLM_classification=YES`, `LLM_personal_funny=NO`), Random Forest models were trained using `quanti_analysis.py` on Tier 2 attributes and confidence [**p22**]. Feature importance scores quantified each attribute's contribution to mismatches, with results tabulated for Reddit and ColBERT (Tables 1, 2, [**p17**]). Mean attribute values for matches vs. mismatches were visualized as bar charts (Figures 9, 10, [**p17**]).

For the random forest, input features were standardized, and NaN values (where present though extremely rare) were imputed using median values to ensure clean training. The Random Forest was configured with 100 estimators and a fixed random seed for reproducibility. To interpret model behavior, feature importances were extracted post-training to identify which factors most strongly predicted ToM-relevant humor.

4.5.2 Qualitative Analysis

For the qualitative analysis, 16 mismatch samples were analyzed and randomly selected across both datasets. This was done in order to understand why certain text was classified as a joke but not personally funny. Cases where text was classified as a non-joke but personally funny did not occur. This allows for the study to probe reasons behind humor classification, personal humor judgments, and jokes for which the LLMs had difficulty. It also allows for extrapolating trends from the quantitative analysis with the qualitative one, allowing for a holistic analysis and finding overarching themes.

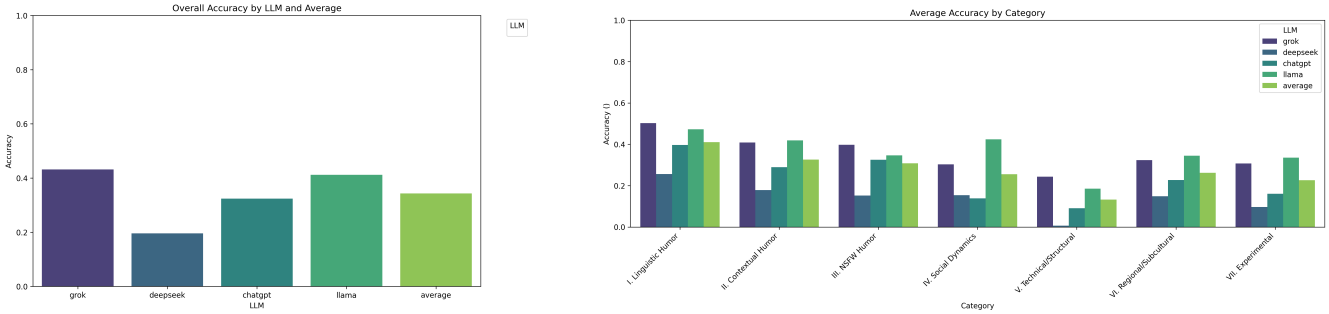
5 Results

5.1 Reddit Data Set Results

The following section has visualizations of general results. For a full table of results, this can be found in the appendix 10.

5.1.1 Tier 1 Results

The general performance of the Reddit dataset across the various LLMs, overall and categories, is as follows: Figure 2 demonstrates the average accuracy of the Reddit Dataset overall and then a



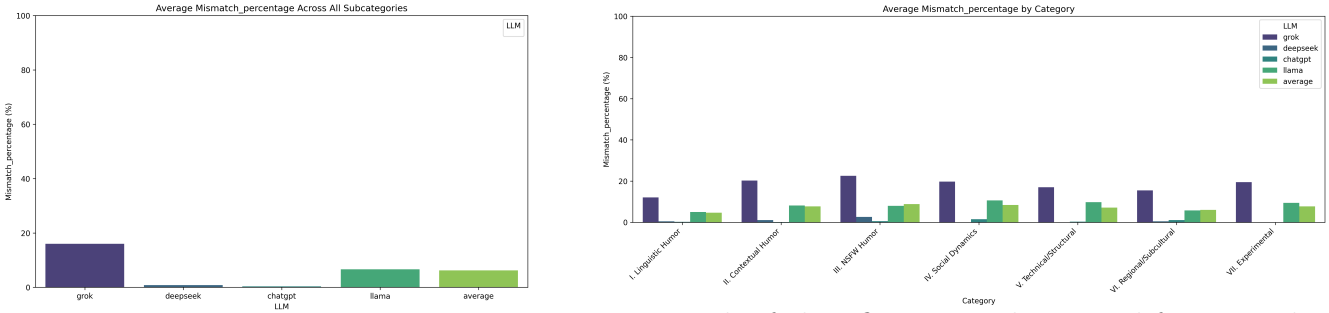
Average Accuracy

Average Accuracy by category

Figure 2: Comparison of LLM accuracy metrics

breakdown by category. Grok, in line with what was hypothesized, had the best performance at 43 percent accuracy, followed by Llama, ChatGPT, and Deepseek at 41, 32, and 19 percent, respectively.

The breakdown by category demonstrates that the various LLMs perform best on linguistic humor, but suffer especially with Structural, experimental, and regional humor. Interestingly enough, on average NSFW humor performs only slightly worse than Linguistic or Contextual Humor. Humor involving social dynamics performed quite poorly, however, not as poorly compared in previous weaknesses. Deepseek seems to have performed nearly abysmally in the technical category.



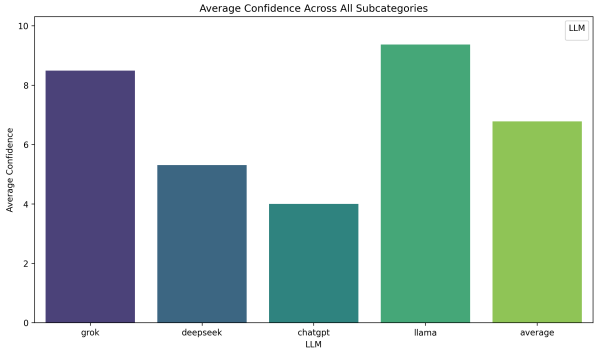
Classification and personal funniness mismatch

Mismatch of classification and personal funniness by category

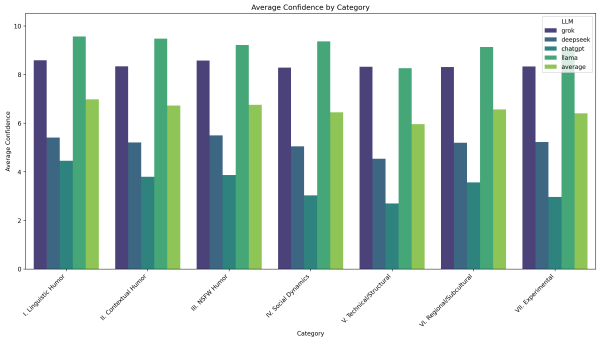
Figure 3: Comparison of LLM classification and personal funniness mismatch metrics

Figure 3 above demonstrates the mismatch between humor classification and the personal funniness classification. For the most part, they seem to mainly align with personal funniness. Deepseek and ChatGPT correlate humor classification and funniness nearly identically, at a mismatch percentage of 0.8 and 0.3, respectively. Grok and Llama have a 15.9 and 6.5 percentage mismatch.

On inspection of the database, mistakes nearly always occur when the LLM finds something to be humorous, but not personally funny. In the breakdown by category, we largely see the same trend taking place as in the overall graph. Mainly changes are minor. With not much surprise, NSFW humor ranks as the highest mismatch percentage for all LLMs, minus Llama. However, the difference between NSFW and Social Dynamics still remains minor.



Average confidence across all subcategories



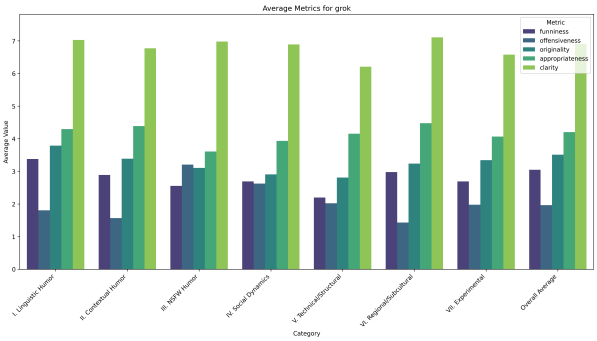
Average confidence by category

Figure 4: Comparison of LLM confidence metrics

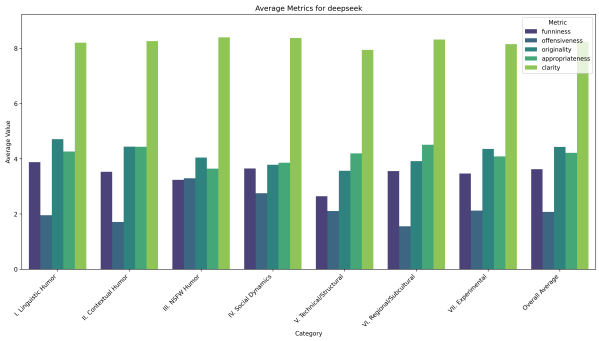
In the figures above, we have a breakdown of the confidence variable, which holds steady across the various categories. Upon inspection of the average confidence across all subcategories, one can see that Llama and Grok are by a large margin the most confident, at 9.3 and 8.4, respectively. Llama and Grok, as per the accuracy chart, are the most accurate, performing roughly at a similar level. ChatGPT, while outperforming Deepseek, demonstrates lower confidence than Deepseek. This maybe can be attributed to stricter moderation policies, training background, and cultural context, which impact their lower confidence. Due to a roughly similar constant confidence, drawing category specific conclusions with our framework is not possible.

5.1.2 Tier 2 Results

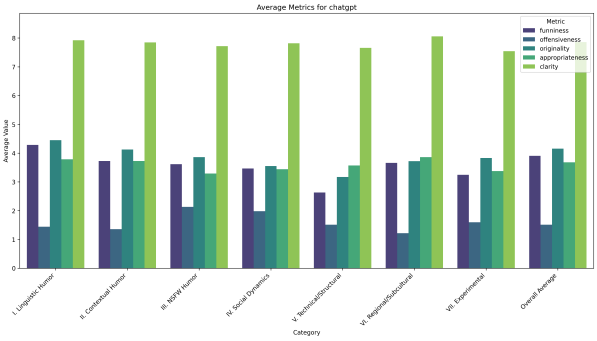
Here we can see the various results as follows for Tier 2. The following figure demonstrates the average metrics for each LLM, with a roughly similar composition for each metric.



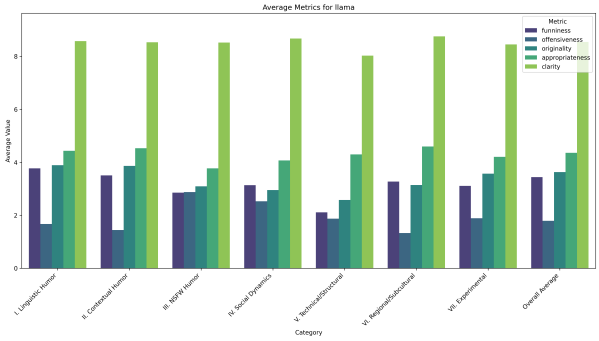
Average metrics by category for Grok



Average metrics by category for DeepSeek



Average metrics by category for ChatGPT



Average metrics by category for Llama

Figure 5: Comparison of average humor metrics by category across LLMs

5.2 ColBERT Data set Results

5.2.1 Tier 1

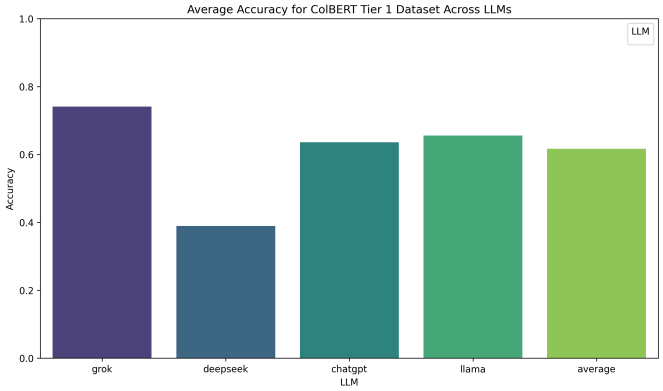
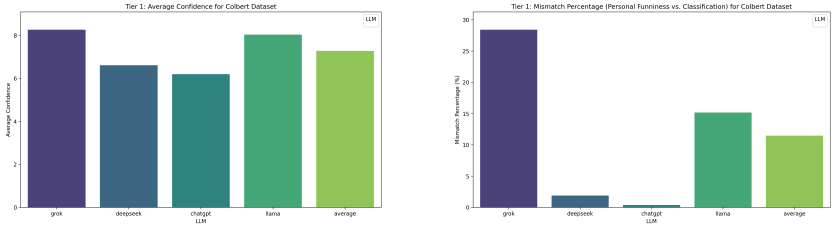


Figure 6: ColBERT Accuracy by LLM

Above, figure six, is the accuracy of the ColBERT dataset. From inspection of the graph, one can see that there is a lesser difference between LLM performance, with the exception of Deepseek. The more "traditional" humor of ColBERT likely plays a large role in this, however previous dynamics are visible as Grok is the best performing, followed by Llama and ChatGPT. On average, the data set achieves a 63 percent accuracy, significantly higher than Reddit's of 0.36.



(a) Confidence (b) Mismatch

Figure 7: Attribute values for Tier 1 LLMs in ColBERT dataset.

The section above, figure 7, demonstrates that confidence in the ColBERT dataset is higher for Deepseek and ChatGPT, but remains roughly identical for Grok and Llama.

5.2.2 Tier 2

From inspection of the attribute Values, we can see some correlations to the Reddit dataset, with Clarity being the highest just as previously seen. A similar case can be viewed for Funniness, offensiveness, and appropriateness. However, for all LLMs the originality of the jokes appear to be higher overall.

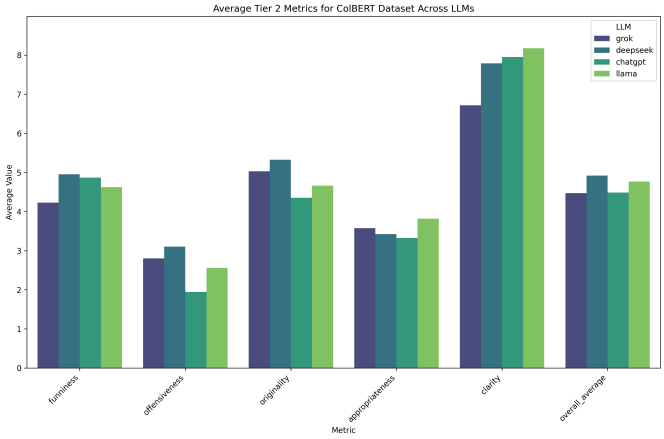


Figure 8: ColBERT Tier 2 Metrics Results

6 Discussion

6.1 Discussion of Inferential Statistics and Preliminary Results

6.1.1 Reddit Dataset

On inspection of the accuracy results of the Reddit data set, while it may seem low, it should be noted that in similar studies, such as on investigating stand-up comedy, LLMs performed with an accuracy of 50 percent, while human annotators lagged behind at 41 percent [36]. Given this context, the underperformance of the various LLMs makes sense, and without human annotators, there is no benchmark to compare them against. Given this and the dataset, one could determine that the various LLMs performed reasonably well.

Moving onto the discussion of category results, within the theoretical framework, structured jokes work best, likely meaning loosely that jokes which are explained by incongruity theory are likely the easiest for LLMs to interpret. On the other hand, regional and NSFW likely perform poorly because they rely on superiority and relief theory. From this perspective, LLMs likely have an easier time picking up on jokes which are funny due to their structure (incongruity theory), such as linguistic or contextual, whereas jokes that may require an emotional aspect are harder for LLMs to detect. This could imply that LLMs, while linguistically very capable, still lag behind in emotional inference or ToM optimization [28]. Furthermore, DeepSeek seems to suffer the most in performance, likely given its distinct training data.

LLM Performance is stronger in more traditional jokes, such as linguistic humor or contextual humor, which consist of puns, entendres, homophones, etc. This is quite likely due to training data. The more niche, offensive, or emotionally demanding the humor gets, the more it requires feelings of relief or superiority; LLMs begin to struggle in detection. This could likely be due to the fact that such jokes are not appearing frequently within training data, and that LLMs lack the proper ToM skills to recognize such frameworks in humor. While the performance of the LLMs may appear to be low, with each having an accuracy rate of under 50 percent, it is unknown if humans would fare better with accuracy, as past studies have indicated that humans and computers alike struggle

in humor classification. Humor is, after all, a subjective classification, with agreement being difficult for human annotators.

However, a few outliers do exist. For instance, performance on the technical category is quite low, while this category is mainly based on incongruity theory, it still likely poses a challenge, as it may present patterns that are difficult for LLMS. This category is composed of misdirection, emoji, the rule of three, and meta humor, which is likely challenging for all LLMs to pick up on. In social dynamics, we see all LLMs suffer, except for Llama. This could likely be down to its community-driven fine-tuning, however, such an explanation is speculative.
From a humor theory perspective, we can see how categories impact and can be explained within the humor theory framework. Linguistic and Contextual humor, rooted in Incongruity Theory's emphasis on unexpected linguistic structures (e.g., puns, homophones, [5]), are readily detected due to LLMs' proficiency in pattern-based semantic analysis, requiring minimal cultural inference [**p7**]. On the other hand, Structural and Experimental humor, despite partial alignment with Incongruity Theory (e.g., misdirection, meta-humor), challenge LLMs with complex or abstract patterns (e.g., emoji absurdism, [**p16**]), as their training data may lack sufficient exposure to such unconventional forms of humor.
Moving on, regional humor, correlated with superiority theory, includes instances of culturally specific mockery (e.g. British Class humor), and as a result, most LLMs perform poorly, indicating struggles with most likely contexts or the humor pathway to a lesser extent. Surprisingly, NSFW did not perform as poorly as expected, noting the LLMs likely did well on BVT.

Moving on to humor mismatches, given the reported results, this could imply that Grok and Llama, to a smaller degree, have a more developed ToM ability, as they can recognize that something may be humorous without it being personally funny to them.

On inspection of the database, mistakes nearly always occur when the LLM finds something to be humorous, but not personally funny. From this perspective, the mismatch percentage can serve as an indicator of behavioral patterns resembling a "taste in humor". The higher mismatch percentage is likely down to differences in training, with the purpose of a personally funny and humor categorization variable to disentangle surface-level humor recognition from judgment. This probes whether LLMs can distinguish between recognizing a joke as humorous and actually endorsing it as amusing.

In the breakdown by category, we largely see the same trend taking place as in the overall graph. Mainly changes are minor. With not much surprise, NSFW humor ranks as the highest mismatch percentage for all LLMs, minus Llama. However, the difference between NSFW and Social Dynamics still remains minor. From a humor theory perspective, this can likely be down to the fact that the violation is not benign, and many jokes may be too offensive for the LLMs to find personally funny.

When inspecting Reddit Tier 2 metrics, interestingly enough, while LLMs may differ in the detection, confidence, and mismatch, these graphs demonstrate that the LLMs roughly interpret each humorous statement in the same manner. This likely means that despite differences in architecture and training, all the LLMs converge on similar perceptions of the content within the Reddit Humor Database. This could likely be down that this is the primary task LLMs are designed for, and hence, while

drawing different conclusions, they excel at the base level of interpretation of sentiment. Such an occurrence is backed up in other studies, as they share similar interpretations due to shared training objectives [11]. Due to roughly similar constant Tier 2 values, drawing category-specific conclusions with our framework is not possible.

6.1.2 ColBERT Dataset

When inspecting each LLMs accuracy on the ColBERT dataset and taking into account its more traditional humor structure. This demonstrates that LLMs struggle in detecting humor that is too niche and online. Take, for example, the best-performing Reddit Category, linguistic humor, which is the most similar overall to the ColBERT Dataset. Despite the improved performance, there is clear room for improvement, as the LLM fails to diagnose several jokes, especially Deepseek. While LLMs may perform better or worse on humor detection depending on humor category, there is still room for improvement, meaning that LLMs, just like humans, still struggle in fully understanding humor or have diverging opinions on humor.

Likely, the former is true, as training data is human-generated and LLMs would come to reflect statistical patterns extracted and content moderation policies imposed. This likely indicates that understanding comes not from replication and statistical patterns but rather something deeper. This is not unlike what Searle argued in his Chinese Room thought experiment, which challenges the idea that syntactic manipulation of symbols (as in LLMs) amounts to true semantic understanding [30].
While LLMs are not an instance of symbolic AI, his argument likely still holds true in this case. This means for LLMs to succeed in holistic humor detection, likely improvements in ToM or other humor components are needed.
For instance, the better performance of the ColBERT dataset is likely due to its more traditional humor, which best mirrors Incongruity Theory, which emphasizes structured linguistic surprises such as puns and one-liners [**p7**, 5]. ColBERT's focus on clear, self-contained humor (e.g., wordplay, [**p11**]) mirrors Reddit's Linguistic category, but likely of a higher quality. This allows for a greater performance by the LLMs, with minimal in-group knowledge needed, minimal ToM or other components in humor detection.

ColBERT's simplicity reduces emotional and cultural demands, narrowing the performance gap among LLMs (except DeepSeek). Deepseek's poor performance could likely be down to a different, bilingual, and unique cultural training. However, the average accuracy of 63 percent indicates that LLMs still face challenges in humor detection. Likely, improving accuracy requires an understanding of group references, ToM abilities, and other capabilities that humans employ in humor recognition.

Moving on, upon inspection of Figure 7, one can see that confidence in the ColBERT dataset is higher for Deepseek and ChatGPT, but remains roughly identical for Grok and Llama. This indicates that the LLMs are likely more comfortable with the ColBERT dataset, signaling that they seem to be more familiar with more traditional forms of humor rather than newer, online, and more niche humorous content. This can indicate a greater confidence of Deepseek and ChatGPT with incongruent-based humor.
The mismatch percentage while following similar proportions to Reddit's, one can see a dramatic

increase in mismatch, especially for Grok and Llama, increasing roughly 10 points for both. Given that mismatches are nearly always jokes that are identified but found not funny, this means that on the whole, the various LLMs find the ColBERT dataset to be less humorous than the Reddit one. Such an increase warrants a further investigation humor mismatch.

Moving onto Tier 2, from inspection of the attribute Values, we can see some correlations to the Reddit dataset, with Clarity being the highest, just as previously seen. A similar case can be viewed for Funniness, offensiveness, and appropriateness. However, for all LLM,s the originality of the jokes appear to be higher overall. Such a case makes sense given that the Reddit dataset is only one source and pulls from several subreddits where jokes may become repetitive or unoriginal. ColBERT, on the other hand, drawing from several corpora, likely offers a more diverse and cleaned corpus of humor, hence such a rating. Just like the Reddit dataset, we can see that the various LLMs roughly interpret humor in the same way. This is a key strength of LLMs.

Such a shared interpretation suggest LLMs have developed a relatively stable heuristic for interpreting text and specifically humor. It highlights that despite consistency in LLM internal scoring mechanisms, differences exist in interpretation.

6.2 Humor Mismatch

While inferential statistics may allow us to see how various LLMs interpret humor, it does not explain how LLMs may interpret humor. By conducting a further statistical analysis, we can see how various variables from Tier 1 and Tier 2 impact the classification given by the various LLMs at hand. By exploring such relations, we can gain insight into why LLMs classify humor as such.

Both Grok and Llama seem to have a higher mismatch percentage of identifying humor, but not finding it personally funny. This happens across both datasets, but is more pronounced on the ColBERT one. To investigate why, we can investigate this from a qualitative and quantitative angle.

6.2.1 Quantitative Analysis

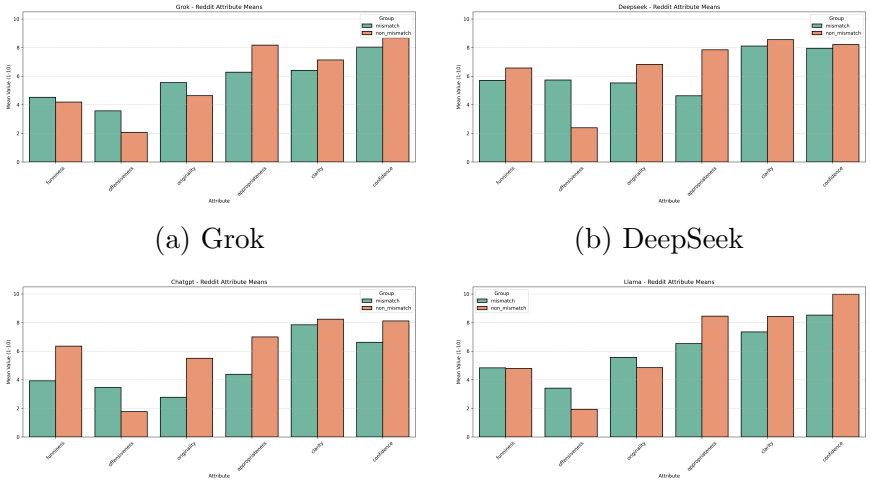
For a qualitative test, this study decided to work only where text is classified as a joke, to best avoid mismatches where the LLM may have misclassified non humor as humor, but still may not find it funny due to its non humorous content. Interestingly, nearly all mismatches for all LLMs are those in which the LLM classifies it as humor, but not personally funny. The other case of misclassification (not hummor but personally funny) account for 0.04-0.01 percent of misclassifications, depending on the dataset and LLM.

To this extent, a random forest ML method was used to determine which variables are likely the best predictors of a mismatch. After determining which factors are the most important, we can view the averages of matched vs mismatched instances, to see if such an effect is positive or not. Below is the table for feature scores, with the highest values being in bold for each LLM.
On inspection of the table, the largest factor for most of the LLMs in mismatch is confidence. For the two most selective LLMs in humor attribution, Llama and Grok, confidence is the dominant

Table 1: Random Forest Feature Importance Scores for Reddit Dataset

LLM	Funniness	Offensiveness	Originality	Appropriateness	Clarity	Confidence
Grok	0.190	0.046	0.036	0.065	0.063	**0.599**
DeepSeek	0.147	**0.303**	0.171	0.233	0.078	0.068
ChatGPT	0.124	0.206	0.124	0.147	0.124	**0.275**
Llama	0.080	0.020	0.022	0.021	0.019	**0.838**

predictor in mismatches. ChatGPT has confidence as a dominant plurality. Deepseek, uniquely, has offensiveness as a mismatch factor, indicating that offensiveness is a primary cause for Deepseek, followed by appropriateness. For ChatGPT, offensiveness is second ranked, not trailing far behind. While such a table is useful as it demonstrates the strength of features, it is important to see how the value of these features impacts mismatching. To this extent, the output of the random forest must be viewed within the lens of comparing descriptive values of mismatches and non-mismatches. When inspecting value averages, we can see that the average confidence, although slight, is lower



(a) Grok (b) DeepSeek

(c) ChatGPT (d) Llama

Figure 9: Mean attribute values for mismatches and non-mismatches across LLMs in Reddit dataset.

on misclassified cases for Grok. Llama displays a more significant difference of nearly one point, scoring nearly perfect on matching cases, but with a one-point difference where mismatches have lower confidence. This means that likely lower confidence, though subtle is still a large component of humor mismatch. For Grok and Llama, difficulty in identification and understanding likely results in making it less personally funny. This could posit that Grok and Llama may have a higher threshold of personal funniness, requiring a larger understanding of the joke or its context that Deepseek or ChatGPT may not demand.

Given the niche context of many Reddit jokes, this can be a cause, as Grok or Llama may recognize humor through superiority or relief theory, but lack the needed information to find said relief or offense humorous. Interestingly enough, Llama and Gro,k while having the highest mismatch rate, have offensiveness play an extremely minimal factor at 0.045 and 0.020, respectively. This

also repeats with appropriateness, demonstrating that Grok and Llama are much more tolerant of offensive humor than Deepseek and ChatGPT. This means that the driver of mismatches is bad quality jokes, or at least jokes which Llama and Grok find to be low quality. Such an explanation is plausible but requires a qualitative investigation.

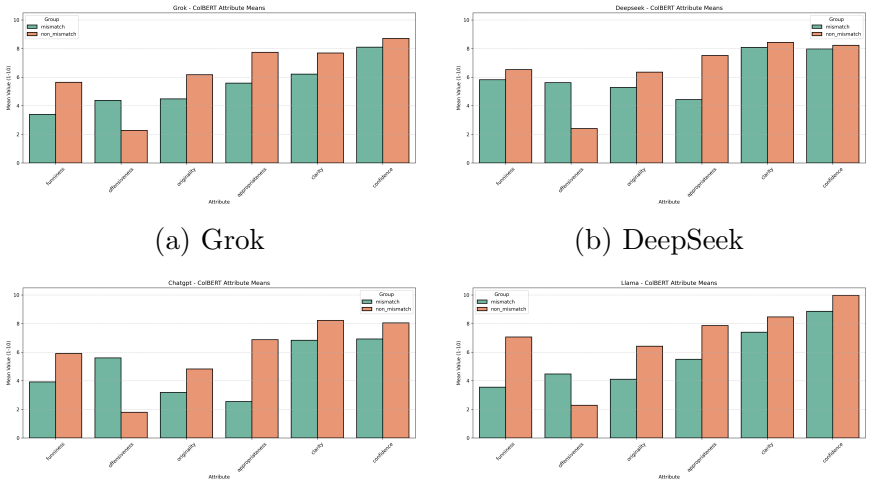
Moreover, ChatGPT, while ranking confidence as the most important category, it remains a plurality, far below the levels of Grok or Llama. Offense comes in as a close second. For Deepseek, offensiveness is the largest driver of mismatching. For both LLMs this can be attributed to a stricter moderation policy. Indeed, Deepseek finds mismatched instances on average 3 times more offensive. Both LLMs find mismatches significantly less appropriate as a whole. This means, unlike Grok or Llama, appropriateness and offense play a large role in the comedic taste of Deepseek and ChatGPT. Both LLMs in the Reddit data set largely prefer safer humor.

While LLMs cannot have a sense of humor, another interesting point is that likely the LLMs do find most mismatches funny on a similar level to non mismatches. In some cases, finding mismatches is funnier. This is likely due to the moderation policy preventing LLMs from finding certain humor as personally funny. This could be similar to hearing an offensive joke as an individual, where while one may find it funny, social norms would make it unacceptable to publicly state so. This is very likely the case for a percentage of mismatches, with the others likely attributed to low quality or unclear jokes.
Now, upon inspection of the ColBERT Database, the random forest highlights the following features.

Table 2: Random Forest Feature Importance Scores for ColBERT Dataset

LLM	**Funniness**	**Offensiveness**	**Originality**	**Appropriateness**	**Clarity**	**Confidence**
Grok	**0.413**	0.062	0.093	0.073	0.125	0.234
DeepSeek	0.121	**0.424**	0.184	0.171	0.028	0.073
ChatGPT	0.116	0.244	0.059	0.113	0.052	**0.416**
Llama	0.358	0.034	0.104	0.011	0.057	**0.435**

When we inspect the mean values for matches and mismatches we can see the following:



(a) Grok (b) DeepSeek

(c) ChatGPT (d) Llama

Figure 10: Mean attribute values for mismatches and non-mismatches across LLMs in ColBERT dataset.

On inspection of the ColBERT Dataset, we see some similar trends to Reddit, offensiveness remains a relatively low indicator of mismatches for Grok and Llama and a high one for Deepseek. However, several shifts in the data set occur as confidence plays a much more minor role, with the importance score halving for Llama (but remaining the most important) and Grok rating Funniness as the most important indicator of mismatch. Given the differences between the ColBERT and Reddit datasets, likely that the Reddit dataset posed a greater challenge for the LLMs as reflected in the inferential statistic section. Funniness remains roughly the same across both datasets for Deepseek and ChatGPT at a second importance of around 0.1. ChatGPT is driven from what is a mostly holistic judgment of the various factors, while Deepseek has a strong aversion to offensiveness and prefers appropriateness.

However, we can notice much larger discrepancies between values, such as funniness. Except for Deepseek, we can observe pronounced differences in funniness between mismatch cases. Clarity largely remains similar to the Reddit dataset, playing a minor secondary role, and on inspection of averages, remains roughly the same across all LLMs and all datasets. This could be that while each LLM likely may not understand fully what occurs, the basic mechanisms of humor, (superiority, linguistic, relief) are likely understood by LLMs and instead prioritize other aspects of humor, such as offensiveness or confidence.

Through a humor theory lens, we can see the shift in ColBERT, while easier for the LLMs to detect humor, it likely lacks the depth some of the Reddit lines have due to ColBERT's focus on self-contained and one-liners. ColBERT lacks the emotional depth or originality for a proper execution of Superiority or Relief Theory-driven humor [**p8**, 24]), leading to pronounced mismatch percentage gaps (Figure 10). This means, unexpectedly, the various LLMs can appreciate and find non-appropriate humor forms as personally funny. For the incongruity theory, Semantic shifts in ColBERT are likely much more predictable and aligned with the training data of the various LLMs, leading to a high mismatch.

Deepseek and ChatGPT have mismatches based primarily on large parts of offense and appropriateness, indicating a struggle with relief or superiority based humor, likely due to a combination of technical ability and moderation policies. Interestingly enough, Deepseek only has a one-point difference in funniness between mismatches and non mismatches, likely indicating that explicit moderation policies drive its 'grumpiness'. ChatGPT, on the other has a significant difference, likely meaning it finds it personally unfunny due to to its training. Even more interestingly, Grok finds mismatched cases as more funny than non mismatched ones, demonstrating mismatches comes from an explicit moderation policy. This could be similar to how certain offensive jokes may be funny to some, but social norms prevent one from saying so publicly.

6.2.2 Qualitative Analysis

However, statistics on 5 metrics alone to explain humor mismatch does not constitute a holistic analysis. With a relatively small sample set of explanations, we can attempt to extrapolate a trend and align it with patterns seen in our quantitative analysis. In the appendix10.0.1 a full breakdown of explanations can be found.

On inspection of mismatch explanations, we see an overlapping pattern, where jokes of a lower quality are found non funny by the LLM. These jokes are often of a low originality, take "Not the brightest chicken" or the "I like my coffee" jokes, which repeat commonly used punchlines, meaning no unexpected shift occurs, and hence such statements, while humor, are not funny. Furthermore, such a joke does not align with the Relief or Superiority theory. Given that while it is a joke, it fails to proper deliver the joke properly in a way outlined by any humor theory. Such a trend repeats with other jokes such as the one about the "the man with his head held high" or "Why buy the horse when you can beat it". For the latter, Grok explicitly states it is not personally funny as it is "too obvious, overused and simply not to their taste". Given this, the shift in tone fails to be unexpected and hence the LLM does not find the joke as funny.

Such a trend remains the same for other humor pathways, such as the "Baltimore Ravens Joke", which requires in-group knowledge of the Baltimore Ravens or the British fish and chips joke. Such Jokes require ingroup knowledge that is largely not clear. Grok states that the raven's joke is forced, likely due to the niche-ness of the Joke. For superiority theory, we can see that the jokes best explained by this framework have poor executions, likely leading to humor mismatch and poor quality jokes overall.

While the main indicator remains to be originality or low-quality jokes, offensiveness can increase the minimum quality for how funny the joke must be to prevent a mismatch. Take for instance the jokes such as "What do you call a nun in a wheelchair? Holy Roller." Where grok states " the joke feels mildly insensitive" in a way that is "crass" and not "clever". Such an explanation implies that the offensive does not make the joke unfunny, but rather the combination of offense and low quality humor. Indeed, more offensive jokes have resulted funny to LLMs, such as "Why do they say all minorities look the same? because once you've seen juan, you've seen jamaul." which Grok classified as personally funny. From a relief perspective, while both may be taboo topics, the second has a better execution of said relief, being unexpected and more offensive than the prior joke. Furthermore, given the likely relevance of the two topics, the first joke falls flat, not meriting

enough offense to cause relief, while the second one does. From this perspective, LLMs can tolerate jokes to the extent that one of the humor theories is properly executed.

From investigating jokes, an overall trend is that they tend to be of a poor quality, likely due to their poor execution, as explained by the various humor theories. While certain characteristics, such as offensiveness, can raise the threshold of what is considered bad humor, it is a secondary factor in driving mismatch. As seen from the qualitative analysis, such a threshold varies from LLM to LLM, with Deepseek and ChatGPT having it raise the threshold raised significantly, while the impact for Grok and Llama appears to be minimal. Even with high offensiveness, the LLMs may rank a joke just as funny as a non-mismatch, but content moderation policies may force it not to find it humorous, as explored in the quantitative section. Given this, LLMs can demonstrate a nuanced approach to humor, in which they can weigh various variables and imposed norms to give an answer if they find a joke personally funny or not.

What we can see is that LLMs personally find jokes unfunny when they are primarily of a bad quality (not original, generally unfunny, etc), and other factors seen in the qualitative and quantitative analysis can influence such a decision. Factors such as offensiveness, and to a lesser extent, lower clarity and lower confidence, reduce the likelihood of an LLM finding humor that it identifies as funny. Given the higher mismatches of Grok and Llama, they likely have a higher bar for what constitutes a minimum quality of humor, while DeepSeek and ChatGPT likely have a lower standard for humor. Unfortunately, the qualitative explanations did not shed light on what this standard may be, as all were critical. Also, given the mismatch percentage being extremely low for ChatGPT and Deepseek, it is likely that Grok and Llama can differentiate humor from Funniness and originality. This means that Grok and Llama can understand something to be humor, or at least an attempt at it, without it having to be funny. Such a phenomenon is likely indicative of a greater level of ToM for Grok and Llama, or at least a different standard of humor. Indeed, for Reddit, Funiness is a weak predictor for Grok and Llama. On the ColBERT data set, funniness is the largest indicator of a mismatch, and also significant for Llama, but other factors combined remain more dominant.

From the theoretical framework, each example is a result of a poor execution of the framework. A shift poorly done (Why did the chicken cross the road) or a BVT, which is not as benign (9/11). Given this, mismatches stem from poor quality jokes, which are determined by the standards of each LLM and explained by the various humor theories.

6.3 Variable Correlations

We can further inspect the inter-correlation of the different metrics to better understand how they impact humor and humor classification.

6.3.1 Classification Predictors.

To determine what variables best correlate with a yes or no classification, a point-biserial correlation test was performed. From inspection of the results below, one can determine the results for classification correlations with Tier 2 attributes and confidence. The tables are inverses of each

other, so we can treat them as one.

We can see that the funniness is the most dominant variable in predicting a classification, except for ChatGPT. Despite this, funniness remains a strong positive indicator for a yes classification with ChatGPT. Funniness is a strong correlator for ChatGPT and Deepseek, but a moderate or weak one for Llama and Grok. This is likely due to a higher grumpiness as previously noted, or the ability of Grok and Llama to separate funniness from humor. Both reasons likely go hand in hand and are interrelated. On the other hand, in the ColBERT dataset, we find that Llama and Grok's correlations with a yes classification of funniness have increased. This could likely be due to the better quality of the jokes and that they are more understandable overall.

However, joke mismatches are higher for all LLMs in the ColBERT dataset, meaning that while funnier overall, a larger subset of what the LLM would classify as not personally funny exists. It should be noted that personal funniness is not a good indicator of humor mismatch.

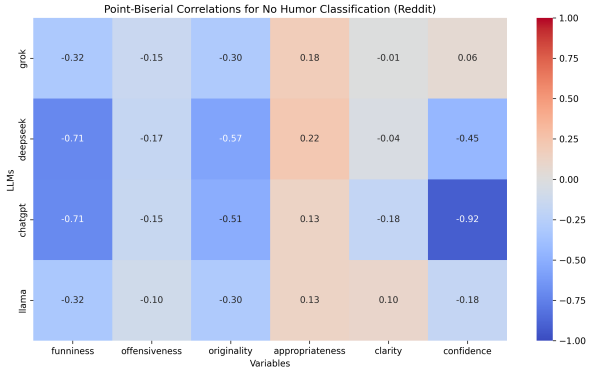


Figure 11: Point-biserial correlation of humor classification predictors without Reddit dataset adjustments.

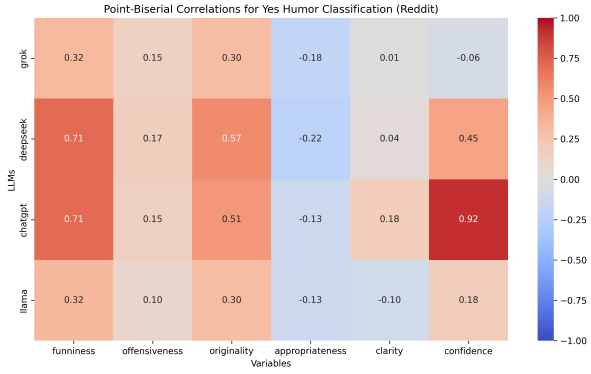


Figure 12: Point-biserial correlation of humor classification predictors with Reddit dataset adjustments.

One noticeable outlier is the confidence predictor in ChatGPT, where the higher the confidence, the more likely it is to give a yes classification on both datasets. On the Reddit dataset, confidence is only a moderate correlator for Deepseek and extremely weak for Grok and Llama. On ColBERT, confidence remains a very strong indicator for ChatGPT, but confidence is now a strong indicator for Llama and a moderate one for Grok. Deepseek remains unchanged. ChatGPT's confidence is reflective of its training, with higher confidence on yes classifications and vice versa. This means ChatGPT likely recognizes the best where it struggles and performs well compared to other LLMs. The other LLMs appear not to have such advanced calibration, but given the increase in correlation on ColBERT, it indicates the LLMs find it easier to perform on. This is reflected in accuracy and means that LLMs likely perform best on incongruity-based humor.

In overall trends, we can see that all variables for Llama and Grok are only weakly correlated with a yes or no classification at best, with few exceptions, likely signaling they are not using the attributes themselves or relating them to a yes or no classification very strongly. Given the previous section, this could be very likely that they can detect humor by discovering 'humor pathways' as

outlined by the humor theories. ChatGPT and Deepseek, do not appear to rely more on sentiment analysis, which may explain lower mismatches and lower accuracy overall. On the other hand, with Grok and Llama, this could signify a better capacity at ToM or just better training data. Whatever the case may be, it demonstrates different approaches to humor detection by various LLMs despite roughly similar sentiment classification in general. This posits that each LLM draws different conclusions and reasons differently despite interpreting humor in a similar manner.

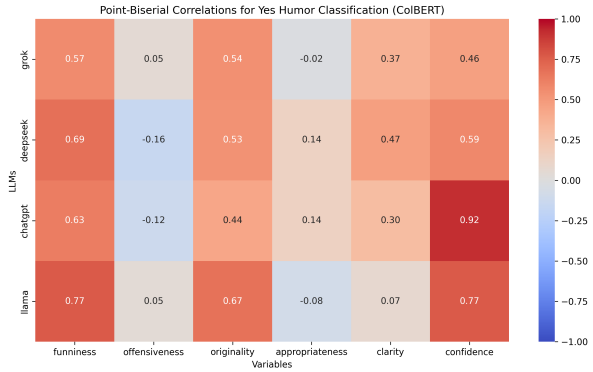


Figure 13: Point-biserial correlation of humor classification predictors with ColBERT dataset adjustments.

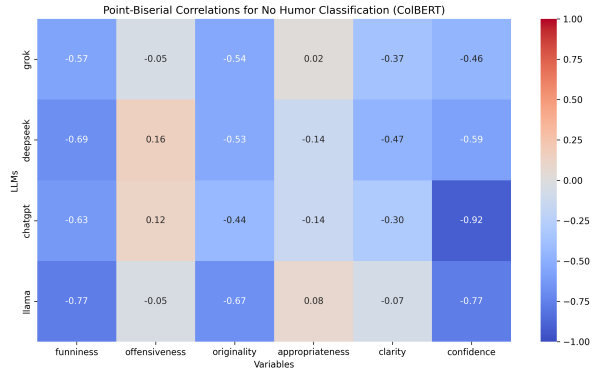


Figure 14: Point-biserial correlation of humor classification predictors without ColBERT dataset adjustments.

From a humor theory lens, the various LLMs, while roughly interpreting sentiment the same, employ different humor detection strategies aligning with: their training, content moderation policy, humor pathways, and other variables [**p6**, 8]. A reliance on the funniness variable indicates a more primitive approach to humor classification, mainly for ChatGPT and Deepseek, while Grok and Llama rely less on individual variables and likely on pathways as outlined in the framework. This is further reinforced by the previous section, where LLMs explain why jokes mismatch due to failing to properly execute a humor pathway, without explicit prompting about any humor theory. In the two datasets, noticeable differences are visible as well, due to their different composition. One can see variable correlations with humor type and pathways. A larger clarity correlation suggests that all LLMs perform best on ColBERT, and by extension incongruent based humor.

6.3.2 High and Low Funniness Predictors

Below, we can see the determining factors in high and low funniness for the ColBERT data set. Interestingly, with the exception of Llama, all LLMs have weak correlations with high funniness; however, within the framework of humor theory, what makes a joke funny is its semantic shifts or other humor pathways. Originality or appropriateness are descriptors, not indicators of humor. However, we can see that for Llama, originality is moderately correlated with high funniness, and confidence weakly so. Likely indicating that Llama finds more original humor funny.

However, more factors correlate with low funniness for ColBERT, mainly originality, which has moderate to strong correlations with lower originality, associating with low funniness. One can also see that confidence plays a role, but only moderately so, as jokes in which the LLM may be unsure are more likely to be unfunny. As discussed previously, low originality can ruin the semantic

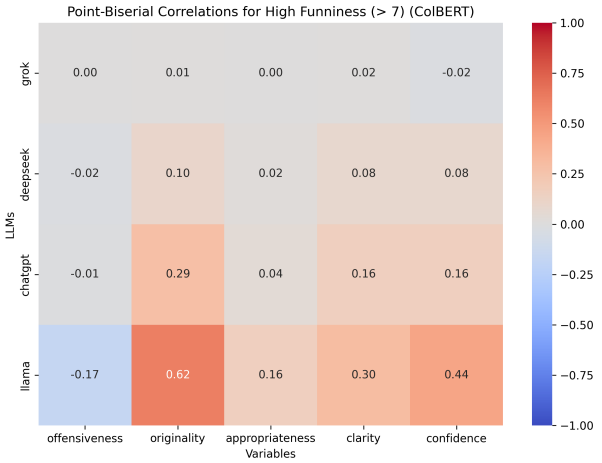


Figure 15: Point-biserial correlation of humor classification predictors for high funniness scores in the ColBERT dataset.

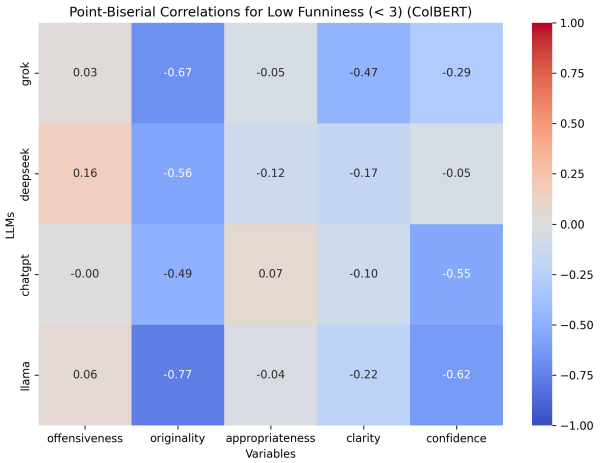


Figure 16: Point-biserial correlation of humor classification predictors for low funniness scores in the ColBERT dataset.

shift and hence make jokes unfunny. For other frameworks, low originality can still impact humor quality, with relief or superiority coming off as overused or stale rather than as each theory posits.

Uniquely, Grok has a moderate correlation for unfunniness with lower clarity. To this extent, unlike other LLMs, unfunny jokes tend to be somewhat less clear to Grok, aligning with its grumpiness and higher humor standards. Grok has the highest cultural and niche knowledge to tackle jokes, so such a bar being raised likely comes from a higher standard for humor or simulating actual human-like grumpiness.

Overall, while the high funniness results matrix does not yield interesting results, the low funniness one does. Originality, likely going in hand with unexpected semantic shifts, means that when the shifts are done in a manner that is expected and overused, all LLMs find it unfunny to a large extent. Given this, it is likely that the primary manner of humor detection by LLMs is the successful execution of one of the humor pathways. The various variables can describe, but probably not indicate, the successful execution of a humor pathway.
For high funniness, we can see the same correlations in Reddit as in ColBERT and draw likely the same conclusions. However, the low funniness correlation matrix changes significantly. Here we can see that originality plays a strong or extremely strong correlation with low funniness, likely indicating that the Reddit dataset is less original overall, resulting in low funniness classifications being unoriginal jokes. One can also see that as appropriateness increases, there is a small to moderate correlation for unfunniness. This suggests that the Reddit dataset and the jokes contained in it are less appropriate as a whole. Interestingly enough, offensiveness contrasts as a weak negative correlation. Likely, this means the jokes within the Reddit dataset are those that are less appropriate as a whole, likely consisting of jokes explained by the Relief or Superiority theory. Combined with the fact that averages are noticeably lower for the Reddit dataset than the ColBERT, this likely means LLMs across the board struggle with Relief and Superiority Theory while doing better with Incongruity theory.

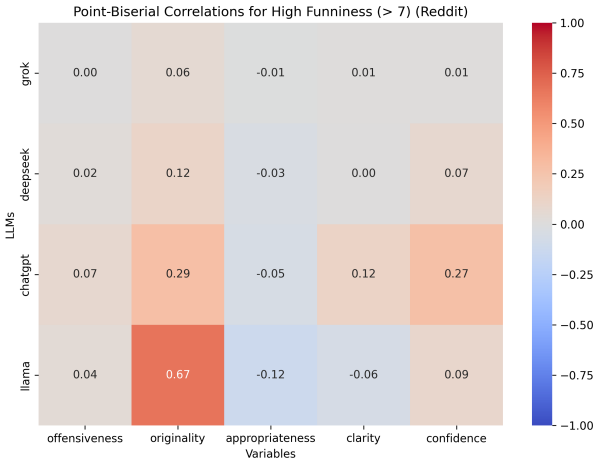


Figure 17: Point-biserial correlation of humor classification predictors for high funniness scores in the Reddit dataset.

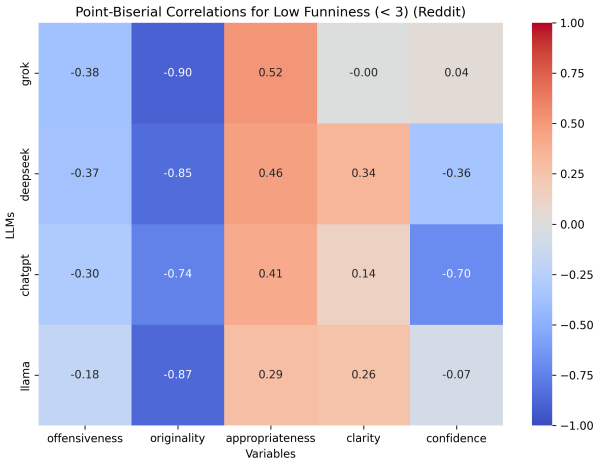


Figure 18: Point-biserial correlation of humor classification predictors for low funniness scores in the Reddit dataset.

Humor theories clarify the correlation patterns for high and low funniness, revealing LLMs' differential engagement with comedic mechanisms.
Reddit's correlations of appropriateness-low funniness and lesser originality than ColBERT indicate its reliance on relief and superiority theory, where LLMs struggle to grasp the needed emotional, social, and ToM context needed to interpret and detect humor[**p8**, 24]. ColBERT, on the other hand, aligns with incongruent humor. From this perspective, the performance of LLMs to detect and interpret humor depends not upon the joke type but rather on the execution of a humor pathway. In this execution, several factors can play into this. The most important ones appear to be joke quality, social context, and LLM ability. Likely, the key to further the development of humor skills for LLMs lies not in training on more jokes explicitly (this would help), but potentially a module or architecture for recognizing joke pathways in combination with having relevant context at hand.

6.3.3 Overall Correlations

Above is the Reddit correlation matrix, which shows the correlation between various variables within the Reddit database. On observation of the matrix, the strongest correlations are between funniness and originality, where the more original a joke, the funnier all 4 LLMs find the joke to be overall. The strongest indicator is that of originality and funniness. Overall the various LLMs found that in jokes that are the most original are the funniest. This is likely due to the fact that when incongruent, they execute the shift better or for relief and superiority cases, said jokes are gracefully executed rather than coming off as crude or crass, as seen in the qualitative analysis.

Another strong indicator is appropriateness and offensiveness, which were predictably very negatively correlated. However, interestingly, the more appropriate a joke is, the various LLMs found it to be less funny and original overall. This reinforces the theory that the primary explainer for an LLM to classify humor and find something personally funny is a successful humor pathway execution. Likely, personally funniness has a higher variance and standards than just simple classification. Conversely,

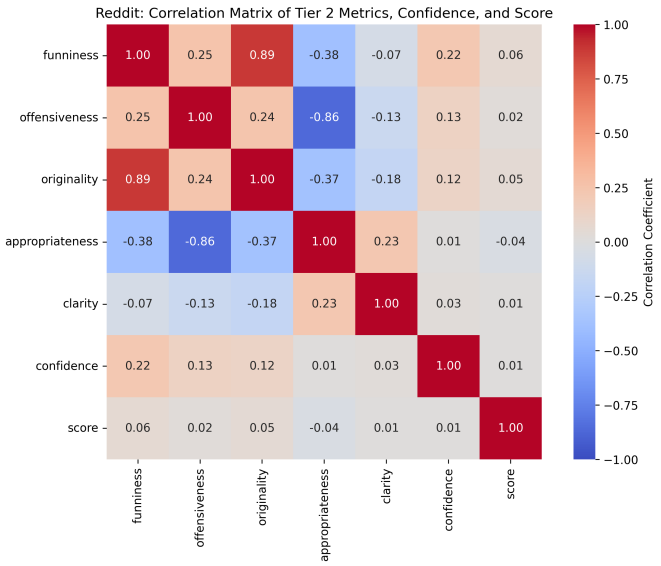


Figure 19: Reddit Variables Correlation Matrix

we see a slight positive correlation between offensiveness with both funniness and originality. This makes sense given the negative correlation between appropriateness and offensiveness.

Overall, the jokes that are slightly more offensive and less appropriate are interpreted as funnier. This is largely expected from the Reddit dataset. Despite this, the primary motivator for funniness is the originality of the joke. Second to this, a weak correlation can be observed is a weak correlation between clarity and appropriateness. Likely more offensive humor, requiring in-group knowledge or other social contexts, is slightly more unclear to the various LLMs.

For the score, we can see no significant correlations, all being essentially negligible correlations. This is likely due to imbalances in the distribution of scores across the various posts.
The matrix above demonstrates similarities and differences with the ColBERT dataset. In the ColBERT dataset, we can see a reduction in the importance of originality correlating with funniness. We see a decrease of about 0.13, which is considerable. Offensiveness passes from being a positive indicator to a negative one, while both weak correlations, which likely indicate that the LLMs find the offensive humor style of Reddit more funny than ColBERT. ColBERT's humor relies on familiar, formulaic patterns, reducing the need for novel twists, which diminishes originality's impact on perceived funniness. Reddit's niche humor (e.g., Regional, NSFW, page 16) demands unique, context-specific content, amplifying originality's role. From the theoretical framework, Incongruity Theory explains this shift. ColBERT's predictable or unoriginal incongruous shifts (e.g., "Holy Roller") require less originality to be recognized as funny, unlike Reddit's unexpected, creative humor (e.g., "Why do they say all minorities look the same…"). LLMs detect these patterns but find ColBERT's humor less engaging due to its conventionality. Take, for instance, the offensiveness of the ColBERT dataset correlating negatively with originality, unlike Reddit.

Moving on, confidence's correlation with funniness nearly doubles, likely because the various LLMs have an easier time identifying and hence finding the humor of ColBERT as humorous. One can

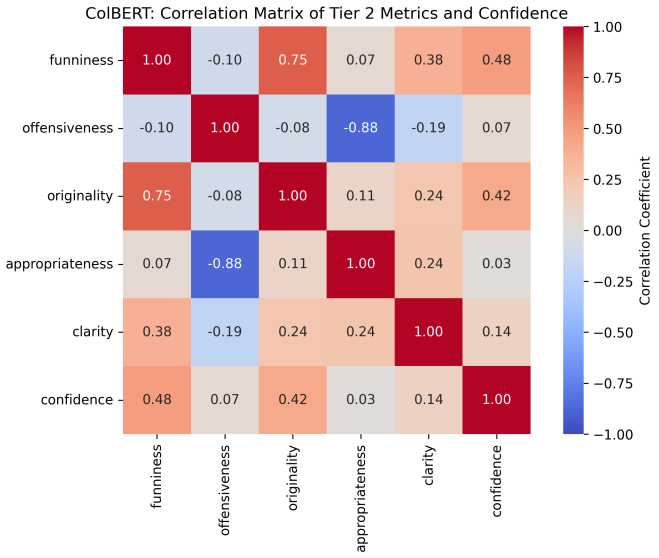


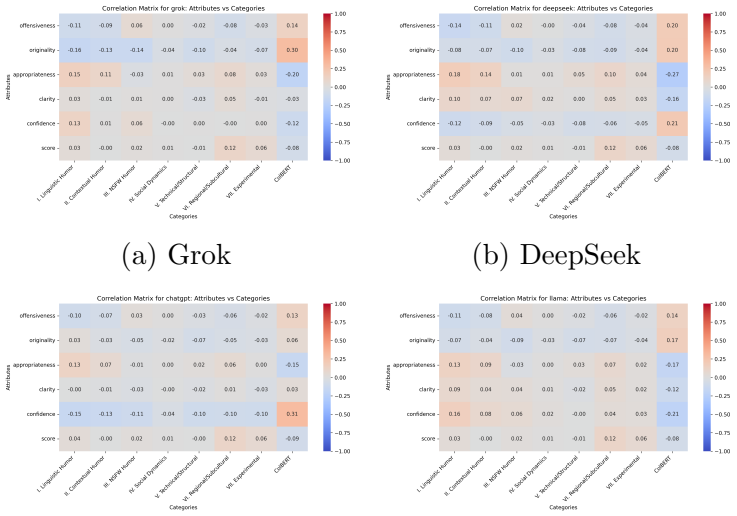
Figure 20: ColBERT Variables Correlation Matrix

also see that a moderate correlation now exists between confidence and originality, nearly tripling from Reddit to ColBERT. Likely indicating that original jokes are much clearer in the ColBERT Dataset. This is likely explained by better performance on incongruent based jokes as previously explained in prior sections.

Other indicators that may demonstrate alignment with superiority or relief theory, such as offensiveness, have weak correlations with funniness. This means that offensiveness, for instance, likely has a limited impact on the funniness ranking for the LLMs as a whole. Even originality, the strongest correlated variable, is likely a descriptor, not a cause for increased funniness. As seen, this means that more funny jokes are original, as originality means a better execution of incongruent humor pathways as previously shown. Even despite this explanation, originality cannot fully be associated with fulfilling the shift required, and several humor theories can likely explain the humor behind one joke. However, ignorance also likely remains the easiest method for LLMs to find jokes humorous, given its simplicity and lack of requirement for more advanced skills and knowledge [**p9**, 22].

6.4 Analysis by Category

The Reddit dataset consisted of several categories and subcategories. The following section will analyze in depth the impact of the categories and subcategories on the performance of the various LLMs. In the following graphs, we can see the correlations.
In the category analysis, we can see that the categories are largely not correlated with any specific metric, likely indicating that the datasets were well balanced for the most part, containing jokes of the same category over a certain distribution. Slight correlations in ColBERT, unlike Reddit's near-absence, suggest its structured humor (e.g., puns, `p11`) aligns marginally with metrics like originality, driven by Incongruity Theory's clear semantic shifts [**p7**, 5]. LLMs' case-by-case humor interpretation, rather than categorical reliance, reflects their pattern recognition strengths but ToM limitations, as they detect Incongruity-based humor yet struggle with Superiority or Relief-driven



(a) Grok (b) DeepSeek

(c) ChatGPT (d) Llama

Figure 21: Correlation matrices of Tier 2 metrics, confidence, and Reddit post score for LLMs.

humor requiring social or emotional context[**p8**, 24].

6.5 Limitations, Weaknesses and Areas of further Improvement

This section outlines the limitations and weaknesses of the study, focusing on methodological constraints, data challenges, and LLM-specific issues, and proposes areas for improvement to enhance future research on humor detection in Large Language Models (LLMs).

6.5.1 Methodological Limitations

Several issues present themselves in the methodology, such as but not limited to the lack of human annotators, small sample size,s or humor thresholds.

One limitation in the methodology was the relatively small sample size used for the qualitative analysis, which was 40 mismatch samples per LLM [**p15**]. While it is enough to capture general trends, it may not capture more interesting examples or all 31 subcategories [**p12**]. Niche humor (e.g: Regional/Subcultural) was underrepresented, limiting ToM insights[8]. This was due to the limited resources at hand to conduct this project, meaning that using crowdsourcing to evaluate a larger number of samples was not an option. Expert labeling would be much more difficult and resource-intensive. Given this, a small sample size was chosen, which new studies may choose to mitigate.

Similar to the previous issue, no human annotators were used to compare the results of the various LLMs to what humans would have scored for the ColBERT and Reddit datasets. This could provide an important reference point to compare LLMs and humans in humor detection. This could allow for a greater insight into Human-AI interaction. However given the aforementioned resource limitations, such an approach was not possible. Furthermore, a deeper investigation of comparing humans and

LLMs would be beyond the scope of this study.

Finally, another methodological limitation was the funniness thresholds used for correlations [**p22**]. While indicative for high and low funniness, such standards are subjective, as the thresholds could have been set at 8¿ and ¡2, respectively. Furthermore, such a heuristic may fail to properly capture data and may leave out important information. Thresholds should likely be driven by data, such as assigning them based on percentiles or other metrics, to more fairly capture trends and information.

6.5.2 Data Set Weakness

Several issues are also present with the current datasets on hand.

For instance, the Reddit dataset, as a consequence of scraping data, is likely of a lower quality. The noise resulting from this may explain the large difference in accuracy between it and ColBERT, 36% vs. ColBERT's 63%, [**p18**]). The extent of this is unknown. Noise in the Reddit dataset may be as follows, but not limited to: posts not properly taken down by moderation, notices to redditors that are not humorous by default. Despite best efforts at dataset cleaning, improved preprocessing could enhance item quality.

Another issue is the fractured nature of the Reddit dataset. The 31 subcategories introduce a high level of variability, with the humor within them likely varying a significant amount in terms of quality or humor pathway execution, among other factors, excluding humor type. Even the distribution of funny or quality jokes may vary significantly from subreddit to subreddit. While this introduces a high level of variability and is valuable in seeing how LLMs react to different types of humor, it may have obscured broader patterns in humor classification and interpretation [**p12**]. This also complicates the aggregation and comparison of results across categories. Future studies may focus on subcategory-level analyses are therefore essential to better understand how specific humor types influence LLM performance. Additionally, the dataset's internal balance across categories remains unverified; future studies may verify this too. This is the case given limitations in time and scope that such an analysis could not be conducted.

One final issue is the simplicity of the datasets. While Reddit offers more advanced humor, ColBERT's complete focus on one-liners is overly simplistic and not representative of humor overall. The Reddit dataset suffers from the same issue to a more limited extent. This issue likely significantly limits the extent to which this humor can be applied outside one-liners and humor in the style of the Reddit and ColBERT datasets.

6.5.3 LLM related issues.

Another issue of limitations was the actual models themselves, which presented several challenges when working with them.

One such limitation was the influence of built-in moderation policies on the outputs of said models.

In particular, ChatGPT and DeepSeek were especially harsh in the enforcement of their policy, which appears to bias their classifications toward "NO" when evaluating NSFW humor content [**p19**]. This could limit their ability to recognize humorous content in taboo or edgy jokes, regardless of structural or context cues, where they otherwise may have. This issue is further compounded as Llama and Grok appear to have more flexible moderation policies, making comparing the models more difficult. These disparities highlight the need to consider moderation frameworks when interpreting model behavior, especially in domains involving sensitive or boundary-pushing content.

Furthermore, a second issue is the calibration of LLM confidence relative to classification outputs. ChatGPT demonstrates a strong correlation between Yes classification and confidence, while Llama exhibits a weaker correlation[**p21**]. These differences very probably stem from variation in how LLMs are trained and fine-tuned for confidence expression, and they introduce interpretive challenges when comparing results across models. A 7 in confidence for Llama may not be directly comparable to a 7 in confidence for Grok, Deepseek, or ChatGPT. Calibration inconsistencies can obscure the reliability of model outputs and complicate any conclusions drawn about their understanding or preferences in the humor detection.

7 Areas of Further Research

Such a study however, given its weaknesses and scope, invites for further areas of research and follow up.

For instance, one possible and promising direction could be conducting a cross-cultural analysis of LLM performance, particularly for models like DeepSeek that are trained on region-specific corpora. Prior results indicate that Deepseek suffers in performance on Western-oriented humor, likely due to limitations in exposure, moderation policy and sociopolitical norms [**p13**]. By comparing its performance against non-Western humor datasets, we could unpack bias and understand LLMs and humor tasks from a non-Western perspective. By then bringing in the humor framework one could clarify how culturally rooted content moderation impacts the model's ability to detect politically incorrect or taboo humor [**p34**] among many possibilities in this research vector.

Another area of further research could be exploring more of the Reddit subcategories and subreddits. Given the scope and constraints of this study, a detailed investigation of subreddits and subcategories was excluded. A future study could do so, exploring subcategories such as regional or dialect-based humor. This would, in theory, reveal how LLMs perform and if such performance is equally capable across various cultural communities, if they present biases, or if certain humor genres systematically elude machine comprehension.

Dynamic moderation represents a third interesting area of possible investigation. The moderation policies of ChatGPT and DeepSeek, disproportionately suppress classifications of NSFW humor [**p19**]. Future research could explore adaptive moderation systems that adjust based on humor type or context, striking a balance between ethical safety and expressive latitude. This, apart from real-life applications, could also provide a more balanced investigation into understanding how LLMs interpret humor and reduce false negatives.

Another and a fourth approach could be benchmarking LLM outputs against human judgments. This could shed insight specific for Human-AI interaction and also provide an important baseline to assess the validity of LLM-generated funniness and humor classifications. Human–AI comparison studies could help determine whether current mismatch metrics reflect actual ToM limitations or simply diverging aesthetic judgments [36]. Such benchmarking would enhance interpretability and rigor in future humor evaluations.

Finally, a distinct and more novel approach could be exploring the performance of LLMs in dynamic humor contexts. This could be in dynamic, interactive environments—such as live chat, interactive comedy, or conversational storytelling—where humor is emergent, layered, and dependent on timing and social cues [**p10**]. These scenarios would test LLMs' real-time ToM capabilities and their ability to follow humor pathways in addition to providing great insight into Human-AI interaction.

8 Conclusion

This bachelor's thesis, conducted at the Leiden Institute of Advanced Computer Science (LIACS), comprehensively and holistically investigated the capacity of Large Language Models (LLMs)—Grok, Llama, ChatGPT, and DeepSeek—to detect, interpret, and explain humorous content. This was done to address a research gap in human-centric AI. By employing a two-Tiered methodology across two datasets, with a humor theory framework for interpretation, this study produced a holistic analysis of LLMs and humor detection and interpretation.

Quantitative results revealed that LLMs are proficient in linguistic and contextual humor, such as one-liners, puns, and wordplay, but suffer with niche, regional, and emotionally driven humor. These humor descriptors, through analysis, were found to correlate with the different humor theories. Incongruity adjacent jokes, such as puns and wordplay, aligning with Incongruity Theory's emphasis on unexpected semantic shifts [3, p16] performed best. Grok achieved the highest accuracy (43% on Reddit, up to 63% on ColBERT), followed closely by Llama (41%), with ChatGPT (32%) and DeepSeek (19%) trailing, underscoring the influence of training data and moderation policies on performance.

On the other hand, LLMs struggled with niche, regional, and emotionally driven humor (e.g., NSFW, social dynamics), which rely on Superiority and Relief theories and require cultural context or emotional inference. This disparity highlights LLMs' strength in pattern-based semantic analysis but exposes limitations in ToM-driven tasks, and niche, emotional contexts where understanding social hierarchies or psychological tension is critical [**.**]

The mismatch investigation demonstrated the humor preferences of each LLM. Generally, across all LLMs, mismatches were driven by low-quality or unoriginal jokes, with offensiveness elevating the quality threshold for humor perception. Such a threshold varied from LLM to LLM. Llama and Grok were willing to tolerate offensive jokes as long as they were funny, that is, executing a Relief or Superiority pathway well. Deepseek primarily prioritized offensiveness, and ChatGPT had a balanced approach across all factors. Notably, Grok and Llama demonstrated greater nuance

in distinguishing humor from personal amusement, suggesting advanced ToM capabilities, while ChatGPT and DeepSeek's stricter moderation policies constrained their engagement with NSFW humor. However, the moderation policies appear to be imposed, not organic, as in many cases, mismatched instances are similar in funniness to non-mismatched instances. This suggests that external content moderation constraints, rather than inherent model preferences, significantly shape LLMs' humor judgments, highlighting the need for more flexible, context-aware moderation strategies to align AI with human-like humor perception. A future investigation can likely determine the trade-offs between flexibility and safety, balancing humor's social benefits with its potential to offend.

The strengths of the study are its holistic approach, theoretical framework, and robust statistical tools, revealing that successful humor detection hinges on the execution of humor pathways rather than joke type alone. However, limitations such as the small qualitative sample size (40 per LLM), lack of human benchmarks, and dataset issues likely limit this study and provide avenues for further research.

Given the understudied nature of the area relating to this study, the real-world applications are several, with various important lessons for advancing human-centric AI. Effective humor comprehension can enhance Human-AI interaction, facilitating a variety of roles from content moderation to virtual assistants. This will allow agents to engage users in natural, socially appropriate methods, facilitating bonding and communication [20]. By delineating LLMs' strengths and weaknesses in humor processing, this thesis underscores the importance of cultural sensitivity and ethical considerations in AI design. For example, DeepSeek's cultural biases, rooted in its Chinese origin, limited its performance on Western-centric humor, suggesting a need for cross-cultural training datasets. Likewise, this could apply to Grok, Llama, or ChatGPT should they expand to non-Western markets. The study's findings inform AI development by highlighting the need for training data that captures diverse, context-rich humor and architectures that emulate ToM, addressing current limitations in emotional and cultural reasoning.

To conclude, this thesis provides a foundation understanding of LLMs and humor detection and interpretation. The study affirmed their ability in linguistic humor while identifying gaps in contextual and emotional comprehension. The research question is answered that LLMs can generally detect and interpret humor, but with varying proficiency and several different caveats, preferences and contributing factors. This was reached using a robust and established humor framework. Future research should prioritize cross-cultural humor analysis, contextual awareness, ToM enhancements, and human-AI comparative studies. By closing the gap between computational prowess and human-like understanding, we can create more human-centric and friendly AI–a critical task given the future prominence and importance of AI. This study, in part, helps understand further how LLMs and humor interact helps pave the way for AI systems that resonate more deeply with the complexities of human communication, fostering richer, more empathetic human-AI interactions.

9 Appendix

10 Results

https://github.com/xocas123/bachelor-thesis

Appendix: Humor Category Definitions

This appendix provides detailed definitions for the humor categories, subcategories, and associated Reddit communities used in the Reddit Humor Dataset, as referenced in Section **??** and Figure 1. Each category and subcategory is defined with its humor characteristics, theoretical alignment, and the full list of subreddits from which content was sourced [15]. The definitions align with the humor theory framework (Section **??**), focusing on Incongruity, Script Opposition, and ToM (ToM) [4, 27, 7].

Linguistic Humor

Linguistic Humor focuses on humor derived from language structure and semantics, emphasizing wordplay and meaning manipulation.

Wordplay Humor based on linguistic ambiguity and structure.

- **Puns**: Jokes exploiting multiple meanings or sounds of words (Incongruity). Subreddits: `r/puns`, `r/dadjokes`, `r/punpatrol`, `r/punny`, `r/badpuns`, `r/cleanjokes`.
- **Homophonic**: Humor using words that sound alike but differ in meaning (Script Opposition). Subreddits: `r/boneappletea`, `r/wordavalanches`, `r/homophones`.
- **Paraprosdokians**: Sentences with unexpected endings, subverting initial interpretations (Incongruity). Subreddits: `r/oneliners`, `r/showerthoughts`, `r/wordplay`.

Semantic Shifts Humor arising from meaning manipulation and ambiguity.

- **Double Entendres**: Phrases with dual meanings, often humorous or suggestive (Script Opposition). Subreddits: `r/dirtyjokes`, `r/wordplay`, `r/standupcomedy`.
- **Malapropisms**: Incorrect word usage creating humorous misinterpretations (Incongruity). Subreddits: `r/boneappletea`, `r/malaphors`, `r/engrish`.

Contextual Humor

Contextual Humor relies on situational or cultural contexts to create comedic effects, often requiring shared knowledge.

Situational Absurdity Humor from illogical or surreal scenarios.

- **Surreal**: Bizarre, dream-like scenarios defying logic (Incongruity). Subreddits: `r/fifthworldproblems`, `r/surrealhumor`, `r/absurdism`.
- **Anti-jokes**: Subvert traditional joke structures with unexpected, literal conclusions (Incongruity). Subreddits: `r/antijokes`, `r/dryhumor`, `r/deadpan`.

Cultural References Humor referencing shared cultural or historical knowledge.

- **Historical Satire**: Jokes mocking historical events or figures (Script Opposition). Subreddits: `r/historymemes`, `r/historyjerk`, `r/ancientmemes`.
- **Pop Culture Parody**: Parodies of modern media, trends, or celebrities (Script Opposition). Subreddits: `r/memes`, `r/terriblefandommemes`, `r/popcultureparody`.

NSFW Humor

NSFW Humor involves mature themes, often pushing social boundaries with potentially offensive content.

Morbid Humor Dark humor focusing on death or suffering.

- **Gallows Humor**: Jokes about death or tragic events (ToM). Subreddits: `r/darkjokes`, `r/blackhumor`, `r/edgydarkjokes`.
- **Terminal Illness**: Humor addressing serious illness or mortality (ToM). Subreddits: `r/toosoon`, `r/morbidhumor`.

Taboo Topics Humor tackling sensitive societal issues.

- **Religious**: Jokes critiquing religious beliefs or practices (ToM). Subreddits: `r/religiousfruitcake`, `r/exchristianmemes`.
- **Political Incorrectness**: Humor intentionally violating social norms (ToM). Subreddits: `r/imgoingtohellforthis`, `r/offensivememes`.

Existential Nihilism Humor reflecting existential despair or cosmic absurdity.

- **Absurdist Dread**: Jokes about life's meaninglessness (Incongruity). Subreddits: `r/absurdism`, `r/existentialmemes`.
- **Cosmic Horror**: Humor involving cosmic fears or the unknown (Incongruity). Subreddits: `r/lovecraftianhumor`, `r/cthulhumemes`.

Social Dynamics

Social Dynamics humor arises from interpersonal relationships and societal structures, often involving power or empathy.

Power Reversals Humor flipping societal power structures.

- **Underdog Humor**: Jokes empowering the disadvantaged (Script Opposition). Subreddits: `r/wholesomememes`, `r/upliftinghumor`.
- **Authority Mockery**: Humor mocking authority figures (ToM). Subreddits: `r/boomershumor`, `r/okboomer`.

Schadenfreude Humor from others' misfortunes or awkwardness.

- **Misfortune**: Jokes about ironic or deserved failures (ToM). Subreddits: `r/leopardsatemyface`, `r/instantkarma`.
- **Cringe Comedy**: Humor from awkward or embarrassing situations (ToM). Subreddits: `r/cringetopia_Two`, `r/sadcringe`.

Technical/Structural

Technical/Structural humor leverages joke structure or visual elements to create comedic effects.

Pattern Interrupts Humor disrupting expected patterns.

- **Rule-of-three Violations**: Breaking the rule-of-three structure for surprise (Incongruity). Subreddits: `r/unexpected`, `r/theydidthemath`.
- **Misdirection**: Leading to an unexpected outcome (Incongruity). Subreddits: `r/nonononoyes`, `r/yesyesyesno`.

Visual Humor Humor using visual or text-based elements.

- **Emoji Absurdism**: Exaggerated emoji use for comedic effect (Incongruity). Subreddits: `r/emojipasta`, `r/deepfriedmemes`.
- **Recursive Meta-Humor**: Jokes about humor or recursion (Script Opposition). Subreddits: `r/recursion`, `r/metamemes`.

Regional/Subcultural

Regional/Subcultural humor reflects specific cultural or subcultural identities, often requiring localized knowledge.

Dialect Humor Humor tied to regional dialects or cultural norms.

- **British Class Humor**: Jokes about British culture and class dynamics (Script Opposition). Subreddits: `r/britishhumor`, `r/casualuk`, `r/britishproblems`.
- **Southern US**: Humor reflecting Southern US culture and stereotypes (Script Opposition). Subreddits: `r/holdmybeer`, `r/holdmyfries`.

Nerd Culture Humor appealing to technical or scientific subcultures.

- **Programming**: Jokes about coding and software development (Incongruity). Subreddits: `r/programmerhumor`, `r/codinghumor`.
- **Science Puns**: Scientific wordplay and humor (Incongruity). Subreddits: `r/sciencememes`, `r/physicsmemes`.

Experimental

Experimental humor explores unconventional or futuristic themes, often challenging LLMs' reasoning.

AI-Specific Challenges Humor testing LLMs' contextual or ethical understanding.

- **Contextual Sarcasm**: Sarcastic humor requiring context (ToM). Subreddits: `r/sarcasm`, `r/fuckyouinparticular`.
- **Ethical Edge Cases**: Humor involving unethical scenarios (ToM). Subreddits: `r/unethicallifeprotips` `r/badphilosophy`.

Temporal Humor Humor involving time discrepancies or futuristic themes.

- **Anachronism**: Jokes mixing historical and modern elements (Script Opposition). Subreddits: `r/historymemes`, `r/timetravelercaught`.
- **Future Shock**: Humor about futuristic concepts or dystopias (Incongruity). Subreddits: `r/cyberpunkmemes`, `r/transhumanismhumor`.

10.0.1 Qualitative mismatch samples

Please refer to the git repository.

bibliography

References

[1] Joost Broekens et al. "Large Language Models are Perceived as More Emotionally Aware than Humans". In: *arXiv preprint arXiv:2309.01664* (2023). URL: https://arxiv.org/abs/2309.01664.

[2] Anthony. *Best 44 Large Language Models (LLMs) in 2025*. Accessed: 2025-05-30. May 2025. URL: https://explodingtopics.com/blog/large-language-models.

[3] Salvatore Attardo. *Humorous Texts: A Semantic and Pragmatic Analysis*. Berlin: Mouton de Gruyter, 2001. ISBN: 978-3-11-017068-9.

[4] Salvatore Attardo. "Humorous Texts: A Semantic and Pragmatic Analysis". In: (2001).

[5] Salvatore Attardo. *Humorous Texts: A Semantic and Pragmatic Analysis*. Berlin: Mouton de Gruyter, 2001. ISBN: 978-3-11-017068-9.

[6] Simon Baron-Cohen. "The Evolution of a Theory of Mind". In: *Annals of the New York Academy of Sciences* 877.1 (1999), pp. 724–737. DOI: 10.1111/j.1749-6632.1999.tb09302.x.

[7] Simon Baron-Cohen. "The Evolution of a Theory of Mind". In: *Annals of the New York Academy of Sciences* 877.1 (1999), pp. 724–737. DOI: 10.1111/j.1749-6632.1999.tb09302.x.

[8] Simon Baron-Cohen. "The Evolution of a Theory of Mind". In: *Annals of the New York Academy of Sciences* 877.1 (1999), pp. 724–737. DOI: 10.1111/j.1749-6632.1999.tb09302.x.

[9] Emily M. Bender and Alexander Koller. "Climbing towards NLU: On meaning, form, and understanding in the age of data". In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics. 2020, pp. 5185–5198.

[10] Vladislav Blinov, Valeria Bolotova-Baranova, and Pavel Braslavski. "Large Dataset and Language Model Fun-Tuning for Humor Recognition". In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Florence, Italy: Association for Computational Linguistics, July 2019, pp. 4027–4032. DOI: 10.18653/v1/P19-1394. URL: https://aclanthology.org/P19-1394.

[11] Tom B. Brown et al. "Language Models are Few-Shot Learners". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 1877–1901. URL: https://papers.nips.cc/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

[12] Tom B. Brown et al. "Language Models are Few-Shot Learners". In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 1877–1901. URL: https://papers.nips.cc/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

[13] Tom B. Brown et al. "Language Models are Few-Shot Learners". In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 1877–1901. URL: https://papers.nips.cc/paper/2020/file/1457c0d6bfcb4967418bfb8ac142f64a-Paper.pdf.

[14] Tanishq Chaudhary, Mayank Goel, and Radhika Mamidi. "Towards conversational humor analysis and design". In: *arXiv preprint arXiv:2103.00536* (2021). URL: https://arxiv.org/abs/2103.00536.

[15] *dataset_creation.py*. Python script for Reddit humor dataset creation. Available in project repository. 2025.

[16] Binwei Feng, Yirong Chen, and Guangfeng Liu. "Humor Detection and Explanation with Large Language Models: A Survey and Future Directions". In: *arXiv preprint arXiv:2310.12345* (2023). URL: https://arxiv.org/abs/2310.12345.

[17] Yifan Gu, Xiang Deng, and Yu Wang. "Humor in AI: Massive Scale Crowd-Sourced Preferences and Benchmarks for Cartoon Captioning". In: *arXiv preprint arXiv:2406.10349* (2024). URL: https://arxiv.org/abs/2406.10349.

[18] Geert Hofstede. "Cultural Dimensions in Management and Planning". In: *Asia Pacific Journal of Management* 1.2 (2004), pp. 81–99. DOI: 10.1007/BF01733682.

[19] Sophie Jentzsch and Kristian Kersting. "ChatGPT is fun, but it is not funny! Humor is still challenging large language models". In: *arXiv preprint arXiv:2306.04563* (2023). URL: https://arxiv.org/abs/2306.04563.

[20] Rod A. Martin and Thomas E. Ford. *The Psychology of Humor: An Integrative Approach*. 2nd ed. London: Academic Press, 2018. ISBN: 978-0-12-812143-6.

[21] A. Peter McGraw and Caleb Warren. "Benign Violations: Making Immoral Behavior Funny". In: *Psychological Science* 21.8 (2010), pp. 1141–1149. DOI: 10.1177/0956797610376073.

[22] A. Peter McGraw and Caleb Warren. "Benign Violations: Making Immoral Behavior Funny". In: *Psychological Science* 21.8 (2010), pp. 1141–1149. DOI: 10.1177/0956797610376073.

[23] John Morreall. *Taking Laughter Seriously*. Albany: State University of New York Press, 1983. ISBN: 978-0-87395-642-0.

[24] John Morreall. *Taking Laughter Seriously*. Albany: State University of New York Press, 1983. ISBN: 978-0-87395-642-0.

[25] Olga Popova and Petra Dadic. "Does AI have a sense of humor? CLEF 2023 Joker tasks 1, 2, and 3: Using BLOOM, GPT, SimpleT5, and more for pun detection, location, interpretation, and translation". In: *CEUR Workshop Proceedings* 3497 (2023), pp. 1888–1908. URL: http://ceur-ws.org/Vol-3497/paper-162.pdf.

[26] Dragomir Radev and Rada Mihalcea. "Computational Humor: A Survey of Recent Advances". In: *AI Magazine* 38.4 (2017), pp. 59–70. DOI: 10.1609/aimag.v38i4.2707.

[27] Victor Raskin. *Semantic Mechanisms of Humor*. Dordrecht: D. Reidel Publishing Company, 1985. ISBN: 978-90-277-1821-1.

[28] Maarten Sap et al. "Neural Theory-of-Mind? On the Limits of Social Intelligence in Large LMs". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022. URL: https://arxiv.org/abs/2210.13312.

[29] John R Searle. "Minds, brains, and programs". In: *Behavioral and Brain Sciences* 3.3 (1980), pp. 417–424.

[30] John R. Searle. "Minds, Brains, and Programs". In: *Behavioral and Brain Sciences* 3.3 (1980), pp. 417–457. DOI: 10.1017/S0140525X00005756.

[31] Nick Statt. "Elon Musk Announces Grok, a 'Rebellious' AI With Few Guardrails". In: *Wired* (Nov. 2023).

[32] DeepSeek Team. *DeepSeek: Advancing Language Modeling for Open-Domain Tasks*. Technical report, DeepSeek AI. 2023. URL: https://www.deepseek.com/research.

[33] xAI Team. *Grok: A Truth-Seeking Language Model for Conversational Reasoning*. Technical report, xAI. 2024. URL: https://x.ai/research/grok.

[34] Hugo Touvron et al. *Llama: Open and Efficient Foundation Language Models*. arXiv preprint arXiv:2302.13971. 2023. URL: https://arxiv.org/abs/2302.13971.

[35] Pedro Henrique Vaz Valois, Adrianna Romanowski, and Kristin Tso. "From punchlines to predictions: A metric to assess LLM performance in identifying humor in stand-up comedy". In: *arXiv preprint arXiv:2504.09049* (2024). URL: https://arxiv.org/abs/2504.09049.

[36] Pedro Henrique Vaz Valois, Adrianna Romanowski, and Kristin Tso. "From punchlines to predictions: A metric to assess LLM performance in identifying humor in stand-up comedy". In: *arXiv preprint arXiv:2504.09049* (2024). URL: https://arxiv.org/abs/2504.09049.

[37] Robert West and Eric Horvitz. "Reverse-engineering satire, or "paper on computational humor accepted despite making serious advances"". In: *Proceedings of the AAAI Conference on Artificial Intelligence* 33 (2019), pp. 9865–9872. DOI: 10.1609/aaai.v33i01.33019865.

[38] Xu Xu and Jennifer Pan. "Government-Imposed Censorship in Large Language Models". In: *Unpublished Manuscript* (2025). URL: https://xu-xu.net/xuxu/llmcensorship.pdf.

[39] Pengcheng Yang, Hao Fang, and Xuemeng Song. "Computational Modeling of Humor Detection with Contextual Understanding". In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. 2022, pp. 5678–5689. URL: https://aclanthology.org/2022.emnlp-main.384.